

EXACT ROBUSTNESS AND COMPLEXITY OF THRESHOLD COMPONENT SCORES

ABSTRACT. We study a fixed-graph score that weights the area of each qualifying active component by a step function of its maximum intensity. For independent rational perturbation boxes, we prove an exact finite-state representation, a lossless stable-component summary with rational witness lifting, and a critical-radius formula with a separate boundary test. Threshold reachability and exact extrema and category radius are fixed-parameter tractable in interaction treewidth, breakpoint count, and the capped cardinality threshold, and are in XP for treewidth alone. An FPT algorithm parameterized jointly by width and breakpoint count for any of these tasks would imply $\text{FPT} = \text{W}[1]$. Threshold reachability remains NP-complete with one breakpoint, any fixed cardinality threshold at least two, and any fixed positive decreasing weight pair; at threshold one, two breakpoints suffice with input-dependent positive weights. With one breakpoint, we classify extrema and category radius for every fixed rational weight pair and every cardinality threshold. Minimum-weight closure explains a maximum–minimum asymmetry at threshold one, while fixed-threshold reductions make the hard regions symmetric at every threshold at least two. For nonnegative nondecreasing weights, all five short-output tasks are polynomial-time solvable and agree with common-offset scoring, equivalently joint translation of all intensity thresholds. Their full score and category images can differ, and explicit image enumeration has a superpolynomial output-size obstruction even in this monotone subfamily.

1. INTRODUCTION

Threshold component scores combine several discontinuous operations: activation of vertices, formation and merger of connected components, a minimum-cardinality test, and selection of a weight from a component maximum. A perturbation can therefore change both the active graph and the multiplier applied to an entire component. We study exact attainable values, extrema, threshold reachability, and the infimal radius permitting a score-category change, together with a separate boundary test, under an independent rational ℓ_∞ box.

One motivation is calcium scoring in computed tomography. The classical Agatston score combines lesion area with a density factor determined by peak attenuation [1]; threshold sensitivity and partial-volume effects are established concerns [13]. The mathematical model here has a fixed finite graph, fixed nonnegative areas, a cardinality threshold, and exact rational intensity data. It is not an acquisition model or a clinical validation. Applying it to an imaging method requires matching that method’s connectivity, lesion-size rule, area convention, and intensity thresholds, and independently justifying the perturbation set.

For nonnegative nondecreasing weights, the short-output questions have a simple answer: the two uniform-offset corners attain the independent-box extrema. Common intensity offsets are exactly equivalent to translating the activation threshold and every density breakpoint together. The two perturbation families consequently have the same category-invariance predicate at every radius, the same infimal failure radius, and the same finite-boundary bit. They need not have the same score or category images. Thus full-image and component-summary questions retain information not supplied by corner extrema, whereas the complexity lower bounds concern the broader weight family.

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For arbitrary signed and nonmonotone weights, we first reduce the continuous-looking rational box to a finite product of threshold states. Contracting each stable-active component then preserves its entire attainable score image when its original cardinality, total area, and attainable maximum-bin interval are retained. Every summary assignment has an explicit rational realization. Category change is governed by finitely many vertex-to-threshold distances, with singleton cells kept separately because the first change need not be attained at its infimum.

The exact algorithms are FPT in interaction width, breakpoint count q , and capped cardinality threshold g_0 , and XP in width alone. A source-graph reduction from Unary Bin Packing rules out FPT in the joint pair (width, q) unless $\text{FPT} = \text{W}[1]$. It shows that g_0 cannot in general be omitted from this tuple; whether q can be omitted while retaining width and g_0 remains open. A fixed-weight Independent Set reduction establishes hardness already for $q = 1$ and gate 2, so width cannot be omitted from the FPT tuple. At gate one, a Multiway Cut reduction establishes hardness for $q = 2$ and input-dependent positive decreasing weights.

At $q = 1$ and cardinality threshold one, the precise fixed-weight distinction is asymmetric. Maximization is NP-hard for every fixed pair $\lambda_0 < \lambda_1 < 0$ and polynomial outside that wedge; minimization is NP-hard for every fixed pair $\lambda_0 > \lambda_1 > 0$ and polynomial outside it. Endpoints and category radius are hard in the union of these wedges and polynomial elsewhere. The polynomial regions include the axes and diagonal. A two-layer minimum-weight closure formulation handles one nonmonotone region, while a vertex-Steiner construction proves the hard cases. For every fixed gate at least two, both extrema are hard in the union of the same two wedges and polynomial elsewhere; the same classification holds when the gate is part of the input. These statements separate proved algorithms from NP-hard regions without assuming an unconditional separation of complexity classes.

1.1. Relation to prior methods. Finite threshold partitions and connectivity dynamic programming are established tools. Tree decompositions organize computation on graphs [4]; logical optimization on bounded clique-width provides a broader framework for fixed formula and label alphabet [5]; and connectivity-state compression can preserve optima with smaller tables [3]. Those methods do not by themselves supply the component summary, rational witness lifting, endpoint conventions, or bit bounds required for this score. Our dynamic program separates numerical objectives from structural states by declaring eventual component weights and checking them when components close.

Connected-component hierarchies such as max-trees vary a threshold on a fixed signal [12]; here intensities vary independently while the score’s thresholds are fixed. Robustness of tree ensembles is also studied [2], but those prediction rules do not contain the component-merger operation. K -best MSO enumeration [8] orders solution values with multiplicities and uses constant-time weight arithmetic; distinct rational score images require separate output-size and bit-cost analysis.

The reductions use classical completeness results of Karp [10], the three-terminal cut theorem of Dahlhaus et al. [6], and the parameterized Unary Bin Packing theorem of Jansen et al. [9]. The closure algorithm is related to Picard’s formulation [11]; its exact rational implementation is described below using polynomial-time network flow [7]. All transformations to the present source model are proved explicitly.

1.2. Organization. Sections 2–3 define the model and its finite-state representation. Sections 4–5 prove exact summary and radius results. Sections 6–7 establish bit-cost upper bounds, and Section 8 gives classical and parameterized lower bounds and the image-size obstruction. Section 9 treats nonnegative monotone weights and common offsets; Section 10 gives the fixed-pair classification. Appendices A and B specify encodings and the scope of formal verification.

2. THE THRESHOLD COMPONENT MODEL

Let $G = (V, E)$ be a fixed finite simple undirected graph, with $n = |V| \geq 1$. Its adjacency relation does not change under perturbation. The input consists of a baseline $x \in \mathbb{Q}^V$, a nonnegative area $a_v \in \mathbb{Q}_{\geq 0}$ at each vertex, an activation threshold $\tau \in \mathbb{Q}$, a strictly increasing list

$$b_1 < \dots < b_q \quad (q \geq 0),$$

weights $\lambda_0, \dots, \lambda_q \in \mathbb{Q}$, and an integer gate $s_0 \geq 1$. There is no sign or monotonicity restriction on the weights and no upper restriction on s_0 .

Define

$$\text{bin}(t) = |\{j : 1 \leq j \leq q, b_j \leq t\}|.$$

For $q > 0$ its cells are $I_0 = (-\infty, b_1) \cap \mathbb{Q}$, $I_j = [b_j, b_{j+1}) \cap \mathbb{Q}$ for $0 < j < q$, and $I_q = [b_q, \infty) \cap \mathbb{Q}$. If $q = 0$, the sole cell is $\mathbb{Q}$. Thus equality at a breakpoint belongs to the upper-index bin. For $z \in \mathbb{Q}^V$ put

$$A(z) = \{v : z_v \geq \tau\}, \quad S_G(z) = \sum_{\substack{C \in \text{cc}(G[A(z)]) \\ |C| \geq s_0}} \left(\sum_{v \in C} a_v \right) \lambda_{\text{bin}(\max_{v \in C} z_v)}. \quad (2.1)$$

For any vertex set W , write $a(W) = \sum_{v \in W} a_v$. Here cc denotes the nonempty maximal connected vertex sets of an induced graph. An empty sum is zero. In particular, activity is determined by the evaluated vector z , and the weight is chosen by the maximum *bin index*, not by the largest weight.

For $e \in \mathbb{Q}_{\geq 0}$ define

$$P_e(x) = \{y \in \mathbb{Q}^V : |y_v - x_v| \leq e \text{ for every } v\}, \quad L_v = x_v - e, \quad U_v = x_v + e.$$

All coordinates may vary independently. We seek the exact attainable set

$$\mathcal{R}_e(x) = \{S_G(y) : y \in P_e(x)\}$$

and its attained minimum and maximum S_e^-, S_e^+ .

For category questions, let $c_1 < \dots < c_m$ be rational cutpoints, with $m \geq 0$, and define

$$\text{cat}(t) = |\{i : c_i \leq t\}|, \quad k_0 = \text{cat}(S_G(x)).$$

The category map is nondecreasing, even when the score is not. Let

$$\text{Bad} = \{e \in \mathbb{Q}_{\geq 0} : \exists y \in P_e(x), \text{cat}(S_G(y)) \neq k_0\}, \quad \rho_{\text{cat}} = \inf \text{Bad}, \quad (2.2)$$

where the infimum is taken after embedding rational radii in $[0, \infty]$, and $\inf \emptyset = \infty$. At a finite answer d , we also require the bit

$$v_d = 1 \iff \forall y \in P_d(x), \text{cat}(S_G(y)) = k_0. \quad (2.3)$$

All results below include edgeless and disconnected graphs, zero areas, $q = 0$, $m = 0$, $s_0 > n$, $e = 0$, and coincident activation and density thresholds.

Figure 1 illustrates how activation, connectivity, the cardinality gate, and the maximum-bin weight interact. It also anticipates the distinction between equal extrema and equal score images.

3. EXACT FINITE REDUCTION AND ATTAINED EXTREMA

The state alphabet is

$$\Sigma_q = \{\perp\} \sqcup \{\mathbf{a}_0, \dots, \mathbf{a}_q\}.$$

Write $\text{st}(z)_v = \perp$ if $z_v < \tau$, and $\text{st}(z)_v = \mathbf{a}_{\text{bin}(z_v)}$ otherwise. For $j = 0$ put $\ell_0 = \tau$; for $j \geq 1$ put $\ell_j = \max(\tau, b_j)$. The active cell is $[\ell_j, b_{j+1}) \cap \mathbb{Q}$ when $j < q$, and $[\ell_q, \infty) \cap \mathbb{Q}$ when $j = q$. Some of these cells can be empty.

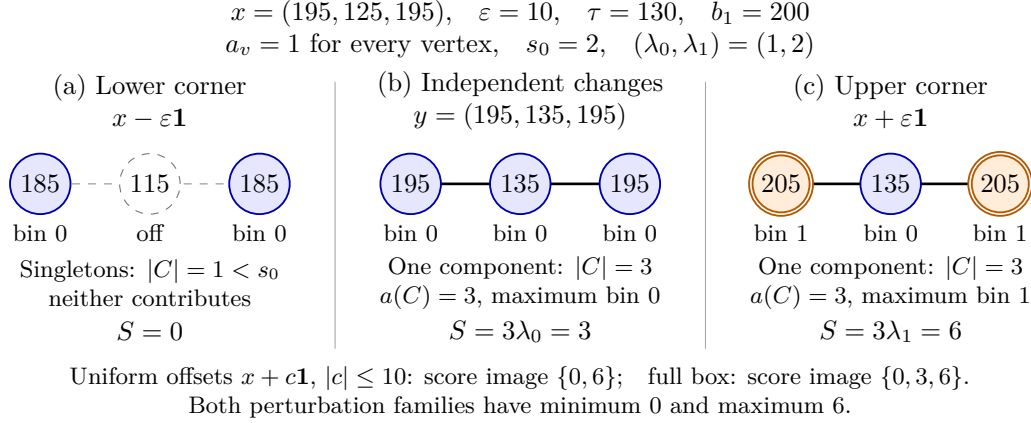


FIGURE 1. A clinical-inspired threshold-component score on a fixed three-vertex path. Vertex labels are intensities; dashed vertices are inactive, and dashed edges belong to the underlying graph but not the active induced graph. Single and double outlines distinguish bins 0 and 1, respectively. At a uniform offset $c = 5$, the middle vertex activates exactly when the endpoints enter bin 1, so score 3 is unattainable by a uniform offset. Independent perturbations can attain it, despite identical extrema. The unit areas and cardinality gate $s_0 = 2$ are schematic model choices, not a clinical scoring protocol.

Definition 3.1. The allowed states at vertex v and radius e are given by

$$\begin{aligned} \perp &\in \text{Allow}_e(v) \iff L_v < \tau, \\ \mathbf{a}_j &\in \text{Allow}_e(v) \quad (j < q) \iff U_v \geq \ell_j, \quad L_v < b_{j+1}, \quad \ell_j < b_{j+1}, \\ \mathbf{a}_q &\in \text{Allow}_e(v) \iff U_v \geq \ell_q. \end{aligned}$$

Let $\Omega_e(x) = \prod_{v \in V} \text{Allow}_e(v)$. For a state assignment s , use its active vertices to form induced components and set

$$F_G(s) = \sum_{\substack{C \in \text{cc}(G[\{v: s_v \neq \perp\}]) \\ |C| \geq s_0}} \left(\sum_{v \in C} a_v \right) \lambda_{\max\{j: s_v = \mathbf{a}_j, v \in C\}}. \quad (3.1)$$

Lemma 3.2 (Exact local realizations). *A state is allowed precisely when its intensity cell meets $[L_v, U_v] \cap \mathbb{Q}$. An allowed inactive state is realized by L_v . An allowed active state $\mathbf{a}_j$ is realized by*

$$r(v, j) = \max(L_v, \ell_j). \quad (3.2)$$

Proof. The inactive cell meets the interval exactly when $L_v < \tau$. For $j < q$, any intersection point is at least both L_v and ℓ_j , at most U_v , and strictly below b_{j+1} ; this gives all three predicates in Definition 3.1. Conversely, those predicates put $r(v, j)$ in $[L_v, U_v]$, at least τ , and strictly below b_{j+1} . If $j \geq 1$, it is at least b_j , so its bin is j ; the same conclusion for $j = 0$ follows from the upper bound. For the top bin there is no upper boundary, and $U_v \geq \ell_q$ is necessary and sufficient. The argument also covers $q = 0$. $\square$

Theorem 3.3 (Finite-state equivalence). *For every instance and $e \geq 0$,*

- (i) *st maps $P_e(x)$ onto $\Omega_e(x)$;*
- (ii) *$S_G(y) = F_G(\text{st}(y))$ for every $y \in P_e(x)$;*
- (iii) *$\mathcal{R}_e(x) = F_G(\Omega_e(x))$, a nonempty finite subset of $\mathbb{Q}$.*

Proof. Lemma 3.2 shows that every legal intensity vector produces allowed states. Conversely, given $s \in \Omega_e(x)$, choose the displayed rational representative separately at each vertex. Independence of the coordinate intervals makes the resulting vector legal and realizes s exactly.

The active vertices of y and $\text{st}(y)$ are identical, hence so are their connected components and areas. The function bin is nondecreasing. On a nonempty finite component, a maximum intensity is attained, and therefore

$$\text{bin}\left(\max_{v \in C} y_v\right) = \max_{v \in C} \text{bin}(y_v).$$

Consequently each summand of (2.1) equals its counterpart in (3.1), including the size gate. No comparison between the weights is used.

The first two assertions give both inclusions of the image equality. The product state space is finite, and it is nonempty because $x \in P_e(x)$. Its score image is therefore nonempty and finite. $\square$

Corollary 3.4 (Exact range endpoints). *The values*

$$S_e^- = \min_{s \in \Omega_e(x)} F_G(s), \quad S_e^+ = \max_{s \in \Omega_e(x)} F_G(s)$$

are attained by legal rational intensity vectors. A minimizing or maximizing state assignment and the representatives in Lemma 3.2 give finite, exact witnesses. The exact largest absolute score error is

$$\max\{S_G(x) - S_e^-, S_e^+ - S_G(x)\}.$$

Proof. A nonempty finite rational set has attained extrema. Lift the corresponding states by Theorem 3.3. The baseline score lies between these extrema, and every score lies in the same interval; its largest distance from the baseline is therefore achieved at an endpoint. $\square$

Remark 3.5. State equality implies score equality, but the converse can fail. The theorem is an equality of score images, not a bijection between rational vectors and states or an identification of state and score equivalence. Nor is $\mathcal{R}_e$ generally an interval. For one isolated vertex with $x = 0$, $e = 1$, $\tau = 0$, $q = 0$, $s_0 = a = 1$, and $\lambda_0 = 2$, the exact image is $\{0, 2\}$.

4. A LOSSLESS STABLE-COMPONENT SUMMARY

Fix e . Partition the vertices into

$$V^+ = \{v : L_v \geq \tau\}, \quad V^- = \{v : U_v < \tau\}, \quad U = \{v : L_v < \tau \leq U_v\}.$$

Let $\mathcal{C}^+ = \text{cc}(G[V^+])$. The interaction graph H_e has the tagged vertex set $U \sqcup \mathcal{C}^+$. It retains the original edges between uncertain vertices and has an edge uC precisely when u has an original neighbor in C . There are no edges between distinct component nodes. Duplicate boundary incidences give one edge.

For $u \in U$ store area $A_u = a_u$, mass $t_u = 1$, and $\beta_u = \text{bin}(U_u)$. When $U \neq \emptyset$, also store $\gamma = \text{bin}(\tau)$. For $C \in \mathcal{C}^+$ store

$$t_C = |C|, \quad A_C = \sum_{v \in C} a_v, \quad \alpha_C = \max_{v \in C} \text{bin}(L_v), \quad \beta_C = \max_{v \in C} \text{bin}(U_v). \quad (4.1)$$

Retain q, s_0 , and all weights. The summary does not store the internal graph of C , its individual intensities or areas, or its vertex support.

A summary assignment r chooses, independently,

$$r_u \in \{\perp\} \cup \{\mathbf{a}_j : \gamma \leq j \leq \beta_u\}, \quad r_C \in \{\mathbf{a}_j : \alpha_C \leq j \leq \beta_C\}. \quad (4.2)$$

Denote this product by $\mathcal{Q}_e$. It has one empty assignment when H_e is empty. Component nodes are always active. For an active component D of H_e put

$$T(D) = \sum_{w \in D} t_w, \quad A(D) = \sum_{w \in D} A_w, \quad J(D, r) = \max\{j : r_w = \mathbf{a}_j, w \in D\},$$

and define its score by

$$E_e(r) = \sum_{\substack{D \in \text{cc}(H_e[\{w:r_w \neq \perp\}]) \\ T(D) \geq s_0}} A(D) \lambda_{J(D,r)}. \quad (4.3)$$

The gate is applied to original mass, not to the number of summary nodes.

Lemma 4.1 (Exact local summary domains). *For $u \in U$, its allowed states are exactly those in (4.2). For every $C \in \mathcal{C}^+$, the attainable maximum bin over its vertices is exactly the integer interval $[\alpha_C, \beta_C]$.*

Proof. For an uncertain vertex, the active portion of its legal interval is $[\tau, U_u]$. A closed rational interval attains precisely the consecutive bins between the bins of its endpoints: intermediate bins are realized by their lower endpoints, and the endpoint bins by the interval endpoints or by (3.2). Thus its active labels range from γ to β_u . Inactivity is possible because $L_u < \tau$.

Every vertex of C is always active and has attainable bins from $\text{bin}(L_v)$ to $\text{bin}(U_v)$. Any component maximum lies between α_C and β_C . Conversely, fix j between these bounds. Some $p \in C$ has $\text{bin}(U_p) \geq j$, while every vertex has $\text{bin}(L_v) \leq \alpha_C \leq j$. Choose intensity $r(p, j)$ at p and L_v at every other vertex. The label j is allowed at p , the other labels do not exceed j , and the maximum is exactly j . $\square$

For $s \in \Omega_e(x)$, define $\pi_e(s)$ by keeping the uncertain labels and replacing the labels within C by their maximum. Lemma 4.1 shows that $\pi_e(s) \in \mathcal{Q}_e$.

Lemma 4.2 (Component correspondence). *For every legal source assignment s , expansion*

$$D \mapsto \text{Exp}(D) = (D \cap U) \cup \bigcup_{C \in D \cap \mathcal{C}^+} C$$

is a bijection from active components of H_e at $\pi_e(s)$ to active components of G at s . Moreover,

$$|\text{Exp}(D)| = T(D), \quad \sum_{v \in \text{Exp}(D)} a_v = A(D), \quad \max_{v \in \text{Exp}(D)} j_v = J(D, \pi_e(s)). \quad (4.4)$$

In particular $F_G(s) = E_e(\pi_e(s))$.

Proof. The active source vertices are exactly all vertices of V^+ together with the selected uncertain vertices. Distinct stable components are disjoint and have no edge between them. Map an active source vertex to itself if uncertain and to its stable component otherwise. Each active source edge either maps to one node or to a summary edge, so a source path maps to a summary walk.

Conversely, each summary edge has a witnessing source edge. Within every stable component, any two attachment vertices can be connected by an active source path. Concatenating these internal paths and the witnessing edges lifts every summary path to a source path. Thus two source vertices are connected if and only if their summary nodes are connected. The fibers partition the active source set, which proves the component bijection and its maximality.

The expanded supports are disjoint unions of singleton uncertain vertices and stable components. Cardinalities and areas therefore add as in (4.4). Taking maxima over the same disjoint union proves the last equality. Each gate and each score summand is consequently preserved. $\square$

Theorem 4.3 (Exact summary and rational lifting). *The projection π_e is surjective, and*

$$\mathcal{R}_e(x) = F_G(\Omega_e(x)) = E_e(\mathcal{Q}_e). \quad (4.5)$$

Every summary assignment has a legal rational realization in the source. Thus summary extrema are exact attained source extrema, and a summary assignment of a different category yields a source perturbation of that category.

Proof. Fix a numbering of V and $r \in \mathcal{Q}_e$. For each stable component with requested label j_C , choose the least vertex p_C satisfying $\text{bin}(U_{p_C}) \geq j_C$; Lemma 4.1 proves that one exists. Define y as follows: on V^- put $y_v = x_v$; at an inactive uncertain vertex put $y_v = L_v$; at an active uncertain vertex of label j put $y_v = r(v, j)$; on a stable component put $y_{p_C} = r(p_C, j_C)$ and $y_v = L_v$ at its other vertices. Each coordinate is rational and legal. The uncertain labels agree with r , and the maximum label on C is j_C . Hence $\pi_e(\text{st}(y)) = r$.

This proves surjectivity. Theorem 3.3 and Lemma 4.2 now give both image equalities. Nonemptiness and finiteness give attained summary extrema, whose displayed lifts attain the corresponding source scores. The score identity also transfers categories. $\square$

Remark 4.4. The standalone summary suffices to compute its full score image. Realizing original intensities additionally requires the source and its numbering. The payload need not determine the baseline score or category; baseline-relative questions additionally require this metadata or the retained source. Stable activation does not fix a density bin: even when $U = \emptyset$, an interval $[\alpha_C, \beta_C]$ can have several elements. Isolated stable components, zero-area components, and components smaller than the gate must be retained. A zero-area component can transmit connectivity; a small component can qualify after uncertain vertices attach.

Proposition 4.5 (Exact size accounting). *Let $u = |U|$, $k = |\mathcal{C}^+|$, and $n_- = |V^-|$. The summary has*

$$N = u + k = n - n_- - \sum_{C \in \mathcal{C}^+} (|C| - 1)$$

nodes and $h = |E_{UU}| + |\mathcal{B}|$ edges, where $\mathcal{B}$ is the set of uncertain-to-component incidences. If E_{--} denotes all source edges touching V^- and E_{++} those with both endpoints in V^+ , then

$$h \leq |E| - |E_{--}| - |E_{++}|, \quad N \leq n.$$

Moreover,

$$|\mathcal{Q}_e| = \prod_{u \in U} (\beta_u - \gamma + 2) \prod_{C \in \mathcal{C}^+} (\beta_C - \alpha_C + 1). \quad (4.6)$$

The node count strictly decreases exactly when $V^- \neq \emptyset$ or a stable component has at least two vertices.

Proof. The partition of V gives the node formula. Every summary uncertain edge is an original edge, and every boundary incidence has at least one witnessing edge not touching V^- and not internal to V^+ . Distinct incidences have distinct witnessing edges. The edge bound follows. Independent local choices give the product, with empty products equal to one; when U is empty no γ is evaluated. The strictness criterion follows directly from the nonnegative subtracted terms. $\square$

Appendix A.1 gives an exact payload length that counts reduced aggregate areas at their actual bit lengths. Structural contraction need not strictly decrease every instance's serialized length.

5. CRITICAL RADII AND EXACT CATEGORY STABILITY

Let

$$\Theta = \{\tau, b_1, \dots, b_q\}, \quad T = \{0\} \cup \{|x_v - \theta| : v \in V, \theta \in \Theta\}.$$

Write its distinct elements as $0 = t_0 < \dots < t_{t-1}$. The radius cells are, in order,

$$\{t_0\}, (t_0, t_1), \{t_1\}, \dots, \{t_{t-1}\}, (t_{t-1}, \infty), \quad (5.1)$$

intersected with $\mathbb{Q}$. There are $2t$ cells and $t \leq 1 + n(q+1)$. A singleton is represented by its point, a bounded open interval by its midpoint, and the tail by $t_{t-1} + 1$. Denote the representative by e_Q and the left endpoint by $\ell(Q)$.

Lemma 5.1 (Constancy within a radius cell). *For any cell Q and $e, f \in Q$, every allowed state set is unchanged:*

$$\text{Allow}_e(v) = \text{Allow}_f(v) \quad (v \in V), \quad \mathcal{R}_e(x) = \mathcal{R}_f(x).$$

Proof. Each radius-dependent predicate in Definition 3.1 reduces to comparisons of the forms

$$x_v - e < \theta \iff e > x_v - \theta, \quad \theta \leq x_v + e \iff e \geq \theta - x_v,$$

for $\theta \in \Theta$. For example, $U_v \geq \max(\tau, b_j)$ is the conjunction of the corresponding two weak comparisons. A negative comparison threshold is automatically satisfied for nonnegative radii. Every nonnegative comparison threshold equals $|x_v - \theta|$ and is in T . Strict and weak comparisons with a point of T are constant on each cell in (5.1). The remaining predicate $\ell_j < b_{j+1}$ does not involve e . Thus all allowed sets are constant. Their products and score images are constant by Theorem 3.3. $\square$

Lemma 5.2 (Extrema decide a category). *At every $e \geq 0$ the following are equivalent:*

- (i) every $y \in P_e(x)$ has category k_0 ;
- (ii) every $s \in \Omega_e(x)$ has $\text{cat}(F_G(s)) = k_0$;
- (iii) $\text{cat}(S_e^-) = \text{cat}(S_e^+) = k_0$.

The same assertions hold with summary assignments and their score in place of source states.

Proof. The image equalities give the first two equivalences, including the summary version. If all scores have category k_0 , so do the attained extrema. Conversely, every attainable score z satisfies $S_e^- \leq z \leq S_e^+$. Monotonicity of cat sandwiches $\text{cat}(z)$ between two copies of k_0 . $\square$

Call Q bad if either extremum at e_Q has category different from k_0 . Let $\mathcal{B}_{\text{rad}}$ be the finite set of left endpoints of bad cells.

Theorem 5.3 (Exact category radius and boundary). *One has*

$$\rho_{\text{cat}} = \begin{cases} \infty, & \mathcal{B}_{\text{rad}} = \emptyset, \\ \min \mathcal{B}_{\text{rad}}, & \mathcal{B}_{\text{rad}} \neq \emptyset. \end{cases} \quad (5.2)$$

In particular every finite radius is in T . If its value is d , its boundary bit is obtained from the separate endpoint-category test at the singleton cell $\{d\}$. At radius zero, category invariance always holds.

Proof. Every nonnegative rational lies in exactly one cell: compare it with the consecutive entries of the sorted finite list T , using the tail beyond its largest entry. By Lemmas 5.1 and 5.2, an entire cell is bad precisely when its representative is bad. Hence Bad is the union of the bad rational cells.

The infimum in the extended nonnegative reals of each cell is its left endpoint. This is immediate for singletons. For an open interval or tail, the left endpoint is a lower bound, and rational points occur arbitrarily close above it; in a bounded interval one may choose such a point below the right endpoint as well. If there are no bad cells, Bad is empty. Otherwise let d be the least bad-cell left endpoint. Every bad radius is at least d , so $d \leq \inf \text{Bad}$. A bad cell with left endpoint d has infimum d and is contained in Bad, giving the reverse inequality. This proves (5.2) without asserting that the left endpoint is bad.

The singleton $\{d\}$ is present in the cell family and Lemma 5.2 decides its invariance exactly. Finally, $|y_v - x_v| \leq 0$ forces $y = x$, proving the zero-radius assertion. $\square$

Example 5.4 (Attained and unattained boundaries). Take a single vertex, $q = 0$, $\tau = 0$, $a = \lambda_0 = s_0 = 1$, and the category cutpoint $c_1 = 1$. For $x = -1$, the baseline score is zero and an active state first becomes possible at $e = 1$. Thus $(\rho_{\text{cat}}, v) = (1, 0)$. For $x = 1$, all intensities remain active at $e = 1$, while inactivity is possible for every $e > 1$. Thus $(\rho_{\text{cat}}, v) = (1, 1)$. For $x = 0$, the answer is $(0, 1)$. These examples show why the infimum and its boundary test are distinct outputs.

6. EXACT TASKS AND POLYNOMIAL BIT BOUNDS

For complexity, vertices are numbered $0, \dots, n-1$ and graphs and rational fields are explicit. We use a deterministic multitape Turing machine with a fixed finite alphabet. Arithmetic, graph traversal, validation, and output writing are all charged. Appendix A specifies canonical self-delimiting natural and reduced-rational codes, complete consumption of input, and the exact task tags. A decision algorithm rejects invalid strings; a function algorithm returns the one-bit error string 0.

At a fixed encoded radius, REACH asks whether $S_G(y) \geq K$ for some $y \in P_e(x)$. The tasks MAX, MIN, and ENDPOINTS return the exact reduced-rational maximum, minimum, and pair of extrema. The task CATEGORYRADIUS has no input radius: it returns ∞ or the pair (d, v_d) from (2.2)–(2.3). The task IMAGE lists all distinct attainable rational values in increasing order. Throughout, REACH denotes the inequality $S_G(y) \geq K$, not exact-score membership $S_G(y) = K$.

Raw inputs contain no decomposition. Supplied inputs contain a decomposition of the constructed H_e , or, for CATEGORYRADIUS, one for each of the $2t$ canonical representatives. Decompositions are checked rather than assumed valid. Let B be the full input length, including any supplied decomposition, and let B_0 denote the corresponding raw length.

Lemma 6.1 (Uniform short-number bounds). *On a valid raw input, every state certificate has polynomial length, every score has $O(B_0 + 1)$ numerator and denominator bits, and every stable-component aggregate area has $O(B_0 + 1)$ bits. Every critical radius and canonical representative also has $O(B_0 + 1)$ bits. The four short-output function tasks have polynomial output length. These score and state bounds are independent of any additional radius at which a category input's assignment happens to be legal.*

Proof. Write $a_i = p_i/d_i$ and $\lambda_j = u_j/v_j$ in reduced form, and put

$$D_a = \prod_i d_i, \quad D_\lambda = \prod_j v_j, \quad D = D_a D_\lambda.$$

The sum of input field bit lengths is at most B_0 up to a fixed coding factor, so these products have $O(B_0 + 1)$ bits. Every score is a sum of at most n terms $a_i \lambda_{j(i)}$, with some vertices omitted. Each term has denominator dividing D , so the whole score does too. Each integer

$$p_i u_{j(i)} \frac{D}{d_i v_{j(i)}}$$

has $O(B_0 + 1)$ bits. Summing at most $n \leq O(B_0)$ terms adds only $O(\log(B_0 + 1))$ bits. Signs and cancellation do not increase the absolute-value upper bound. Reduction of the fraction cannot increase its numerator or denominator. The same argument with D_a proves the area claim.

A state consists of n labels from $q+2$ possibilities. Its canonical code has $O((B_0 + 1) \log(B_0 + 2))$ bits because $n, q \leq O(B_0)$. Critical distances require one rational subtraction; midpoints and the tail representative require a bounded number of arithmetic operations, so their lengths remain linear. Extrema are attained scores, and a finite category radius is a critical distance; adding a boundary bit preserves polynomial length. The score and certificate expressions contain no radius, which proves the final assertion. For IMAGE this proves a per-value bound only. $\square$

Theorem 6.2 (Polynomial validation, preprocessing, and evaluation). *There are total polynomial-bit-time algorithms that:*

- (i) *validate the exact raw or supplied task language, including all rational canonicity and decomposition conditions;*
- (ii) *construct the canonical summary at the input radius, or all canonical critical representatives and summaries for a category input, together with explicit source-to-summary maps;*
- (iii) *evaluate $F_G(s)$ from a source input and a syntactically valid state certificate, and test legality if the radius is supplied and charged.*

The preprocessing output and working space are polynomially bounded in B_0 for raw inputs.

Proof. Natural codes are decoded by scanning a unary length prefix followed by the prescribed binary suffix. Truncation rejects immediately. A decoded integer is retained in binary. Before looping over any advertised list length, compare it with the remaining input length divided by a lower bound of one bit per record. An impossible count rejects without allocating that list. Thus a malformed exponentially large count cannot initiate exponentially many iterations. The gate s_0 is compared in binary and never used as a parsing-loop bound.

School arithmetic gives polynomial-time addition, subtraction, multiplication, and division on binary integers. Euclid’s algorithm has only linearly many division stages in the bit length: after at most two successive divisions the larger remainder is at most half the previous larger argument. Hence rational normalization by gcd and exact comparison by cross multiplication are polynomial. Check nonnegative areas and radii, positive n and s_0 , ordered breakpoints and categories, sorted distinct legal edge pairs, and reduced rationals with a unique zero. No invalid list is repaired by sorting. Reject unsupported headers, tags, modes, trailing data, and truncated fields.

For a fixed radius compute L_v, U_v , classify vertices, and find the stable-active components by graph traversal. Number uncertain vertices in source order and components by their least source vertex. Compute (4.1) by exact sums and comparisons. Scan source edges to construct and sort the output summary edge list, removing duplicate boundary incidences. Record the map sending a stable-inactive vertex to zero and every other vertex to one plus its summary-node index. This constructs precisely the summary proved correct in Section 4. Output summaries and maps are length-prefixed using Appendix A; sorting here concerns a newly generated output. For category inputs, sort and deduplicate at most $1 + n(q + 1)$ distances and produce all $2t$ representatives, including every singleton.

A supplied decomposition has a nonempty finite tree of bags. Check that its listed tree is simple, connected, and has one fewer edge than nodes. Check coverage of all summary vertices and edges and connectedness of the occurrence subtree of every summary vertex. These conditions exactly characterize a tree decomposition. For category input also check the exact number and canonical order of the decomposition records. Counts and occurrences are bounded by the actual input length, while the constructed graph is polynomial in that length; repeated scans therefore suffice in polynomial time.

For state evaluation, parse exactly n legal alphabet labels. Traverse the active induced graph, count each component’s original vertices, sum its areas, find its maximum bin index, compare its size with s_0 , and sum the gated contributions. This is (3.1). If a radius is provided, test Definition 3.1 at every coordinate. Evaluation alone makes no feasibility claim at an unprovided radius.

All loops described above range over polynomially many explicit records or pairs of records, with fixed nesting depth. Lemma 6.1 bounds all arithmetic fields. A summary has at most n nodes and $|E|$ edges; its payload and source map use $O((B_0 + 1)^2)$ bits. There are $O((B_0 + 1)^2)$ category records, so length-prefixed summaries and maps use $O((B_0 + 1)^4)$ bits, including length prefixes. Sequential multitape scans may increase a fixed polynomial exponent but do not introduce parameter-dependent exponents. Storing the complete output already gives a polynomial space bound. Canonical primitive codes and ordered lists also give both encoding round trips. $\square$

Corollary 6.3. *Raw REACH is in NP. The same holds for the supplied-decomposition decision language.*

Proof. The certificate is the canonical state code. Validate the task and certificate, check every local allowed-state predicate at the encoded radius, evaluate $F_G(s)$, and compare with K . Length and time are polynomial by Lemma 6.1 and Theorem 6.2. Theorem 3.3 proves that such a certificate exists exactly when a rational perturbation of score at least K exists. In supplied mode first validate the suffix. Invalid strings have no accepting certificate. $\square$

7. EXACT ALGORITHMS ON TREE DECOMPOSITIONS

For a decomposition $\mathcal{D} = (T, (X_t))$ use the clipped width

$$d(\mathcal{D}) = \max_t \max(|X_t| - 1, 0), \quad \text{tw}_0(H) = \min_{\mathcal{D}} d(\mathcal{D}).$$

The decomposition tree is nonempty and empty bags are allowed. Thus tw_0 is zero for both the empty graph and nonempty edgeless graphs. Define

$$w_e = \text{tw}_0(H_e), \quad w_* = \max_Q \text{tw}_0(H_{e_Q}), \quad g_0 = \min(s_0, n + 1). \quad (7.1)$$

For supplied tasks use d_e and $d_* = \max_Q d(\mathcal{D}_Q)$ instead. These are widths of the actual supplied records, not assertions of optimality.

The whole-radius width. If $0 \leq e \leq f$, then H_e is a minor of H_f . Delete from H_f the representations of vertices in V_e^- and contract the connected representations of each component of $G[V_e^+]$. Since $V_f^+ \subseteq V_e^+$, no stable node of H_f meets two such components, and their representations remain connected by the source paths. This produces H_e , after removing loops and duplicate edges. Consequently w_e is nondecreasing in e . Every H_e is a minor of G , while the canonical tail representative exceeds every $|x_v - \tau|$ and makes every vertex uncertain. There $H_e = G$, so $w_* = \text{tw}_0(G)$. This concerns optimal raw widths; d_* remains the maximum width of the actual supplied records.

A task is FPT in a parameter tuple p if one total deterministic machine solves every bitstring in time $f(p)(B + 1)^c$ for a total computable f and a constant c fixed before all inputs. Invalid inputs have parameter zero. XP in width means a bound $f(w)(B + 1)^{g(w)}$ with total computable f, g .

7.1. The finite interface and its objective. If $s_0 > n$, all scores at every radius are zero. All short-output tasks are then solved after validation without iterating to s_0 . Otherwise write $s = s_0 = g_0$ and use

$$\text{cap}_s(t) = \min(t, s), \quad \text{cap}_s(a + b) = \text{cap}_s(\text{cap}_s(a) + \text{cap}_s(b)). \quad (7.2)$$

The identity follows by separating the cases $a \geq s$, $b \geq s$, and both below s . In particular $t \geq s$ exactly when its cap is s .

Normalize a supplied decomposition to an edge-introducing nice decomposition with empty root and leaves. Along a parent-child connection, first remove the vertices outside the bag intersection and then add the other endpoint's vertices. Replace branching by binary equal-bag joins. Assign each graph edge to one containing bag and insert one equal-bag edge-introduction event there. The resulting decomposition has the same width and $O((r + 1)(d + 2) + h)$ nodes for r original bags and h edges. Every vertex has a unique topmost forget event. Every incident edge is introduced before that event, and at a join the forgotten vertex sets in the children are disjoint.

Fix a summary and a bag X . A structural state contains:

- (i) for each bag vertex, either inactivity or an own allowed label $j \in \{0, \dots, q\}$, a declared eventual component maximum $J \in \{j, \dots, q\}$, and a declared gate outcome $g \in \{0, 1\}$;
- (ii) a partition of the active bag vertices describing connectivity in the already processed active graph; vertices in one block have the same (J, g) ;

- (iii) for each block, the capped original mass of its forgotten vertices and a bit recording whether one of those forgotten vertices has own label exactly J .

An always-active component node may never be assigned inactivity. Bag vertices have not yet contributed mass, witness bits, or objective. All already closed components must have their declared maximum and gate outcome verified. The value of a partial realization is

$$W = \sum_{\substack{v \text{ forgotten} \\ v \text{ active}}} g_v A_v \lambda_{J_v}. \quad (7.3)$$

For each structural state store one maximum such value, or an infeasible marker; a second copy with minimum gives the lower endpoint. Partial area sums are absent from the state key. A guessed eventual multiplier makes each forgotten vertex's contribution additive, while later checks enforce that guess.

An attaining assignment is recovered by retaining one predecessor transition for each selected extremal entry and backtracking from the root. The recovered own labels lift to source intensities by Theorem 4.3. Transition indices and rational witnesses have polynomial bit length relative to the existing table and source bounds, so this does not change the FPT or XP classification.

7.2. Transitions and proof of exactness. The transitions are as follows.

- At an empty leaf, the sole empty state has value zero.
- At vertex introduction, choose allowed local data. An active vertex forms a singleton partition block with mass zero and witness bit zero. No objective is charged.
- At edge introduction, do nothing if an endpoint is inactive. Otherwise require equal (J, g) at the two endpoints and merge their blocks. Sum and cap their masses and take the disjunction of their witness bits. If they already share a block, leave its annotations unchanged.
- At forgetting an inactive vertex, remove it. At forgetting an active vertex v , add t_v to the block mass, cap it, set the witness bit if $j_v = J_v$, and add $g_v A_v \lambda_{J_v}$ to the objective. If this removes the block's last bag vertex, require a true witness bit and require $g_v = 1$ exactly when the capped mass is s , then delete the block. Otherwise retain the resulting block.
- At a join, require identical local data on the two copies of the bag. Take the equivalence closure of the two child partitions. For each resulting block, sum each contained child block's mass once and cap the result; take the disjunction of all its witness bits. Require consistency of (J, g) across the block. Add the two numerical values.

For each parent state optimize over all compatible choices. Equal-bag identity nodes simply copy states.

Theorem 7.1 (Correctness of the extremum dynamic program). *At the empty root, the maximum and minimum values computed by these transitions are exactly S_e^+ and S_e^- .*

Proof. We prove by induction that a table entry optimizes exactly (7.3) over partial assignments with its declared interface and validated closed components. At a leaf there is only the empty assignment. A newly introduced vertex has no processed incident edge and no forgotten contribution, so the introduction transition enumerates exactly its possible local assignments. An introduced active edge forces the eventual data of its endpoints to agree and connects precisely their two current components; a within-component edge changes neither mass nor witnesses.

Forgetting a vertex counts and charges that vertex exactly once. If the last bag vertex of a processed component disappears, the component cannot later acquire an edge: every edge incident to a forgotten vertex has already been introduced, and the component no longer meets the bag separating the processed graph from the remainder. It is therefore final. Every own label is at most the declared J , so a true witness bit is equivalent to actual maximum J . Equation (7.2) makes the mass test equivalent to the original gate. The closure tests are thus necessary and sufficient.

At a join the processed child graphs intersect only in the bag. Their actual connectivity on the bag is the equivalence closure of their partitions. Their forgotten vertices are disjoint, so masses, witnesses, and objective contributions combine exactly as prescribed. No bag vertex has yet been counted, and no subtraction from a capped mass is needed. A child block is included once even if several of its vertices lie in the same resulting block.

Conversely, every partial realization restricts to child realizations and a listed transition. Any compatible child realizations can be glued: all information on which later compatibility depends is in the structural state. The resulting recurrence on sets of numerical values consists of unions, translations, and sums of independent child sets. For finite nonempty rational sets,

$$\max(A + c) = \max A + c, \quad \max(A + B) = \max A + \max B,$$

with analogous minimum identities. Therefore keeping one extremum per state is exact, including for signed weights.

At the empty root all components have closed. Within each component the charged terms sum to its area times its verified maximum-bin weight when it meets the gate, and to zero otherwise. This is (4.3). Every summary assignment yields an accepting derivation by declaring its actual final maximum and gate outcome; every accepting derivation yields a legal assignment with that score. Theorem 4.3 transfers the resulting extrema to the source. At least one accepting assignment exists, including for an empty summary. $\square$

Lemma 7.2 (State and bit complexity). *For width d and $s_0 \leq n$, the number of structural states per bag is at most*

$$Z(d, q, g_0) = [4(d + 2)(q + 2)^2(g_0 + 1)]^{d+1}. \quad (7.4)$$

The supplied-decomposition extremum algorithms take

$$C(Z(d, q, g_0) + 1)^6(B + 1)^c \quad (7.5)$$

bit steps for absolute constants C, c .

Proof. For a bag of size $b \leq d + 1$, there are at most $(1 + 2(q + 1)^2)^b$ local assignments, at most $\max(1, b^b)$ partitions, and at most $[2(g_0 + 1)]^b$ annotation choices. Their product is bounded by (7.4), which also covers an empty bag. Enumerate partitions by representative arrays and test them, so the state bound is uniform and effective.

All transitions can be generated by explicit scans; joins try state pairs, compute a finite equivalence closure, and locate the resulting state by scanning the table. A fixed power such as six of $Z + 1$ covers these operations and their tape-access overhead. The number of nice nodes is polynomial in the supplied length.

Every partial objective is a sum over disjoint source vertices, each multiplied once by some λ_j or zero. The proof of Lemma 6.1 therefore bounds its numerator and denominator by $O(B + 1)$ bits. Optimizing an entry selects one realized partial value; it does not sum competing derivations. Arithmetic and rational reduction have polynomial cost. State fields and mass calculations use binary integers, and no iteration to a binary magnitude is hidden. These observations give (7.5) with a fixed exponent. $\square$

7.3. Constructing a decomposition for raw input. Decomposition construction is included in the raw-input running time.

Lemma 7.3 (A bounded-width decomposition can be found). *There is a uniform algorithm that, on an N -vertex graph of clipped treewidth w , constructs a decomposition of width at most $4w + 4$ in time $f(w)(N + 1)^c$ for a computable f and an absolute c . Its output has polynomial length with an exponent independent of w .*

Proof. Treat the empty graph with one empty bag and a nonempty edgeless graph with singleton bags. Otherwise apply [4, Theorem 2] for $k = 1, 2, \dots$ until a decomposition is returned. That theorem returns width at most $3k + 4$ or correctly reports treewidth greater than k , in $O(c_0^k N \log N)$ RAM time for an absolute c_0 . It succeeds by $k = w$, so the width is at most $3w + 4 \leq 4w + 4$. Component decompositions can be joined for disconnected graphs.

Implement the finite graph computation on a multitape machine using explicit records and sequential table lookup. A polynomial overhead in its operation and storage bounds and index lengths is absorbed in $f(w)(N + 1)^c$; preservation of the RAM time bound is not asserted.

To bound the serialized output independently of the parameter factor, root the returned decomposition and repeatedly contract an edge whose child bag is contained in its parent, retaining the parent bag. This preserves width, coverage, and connected vertex occurrences. In the resulting tree, each nonroot bag contains a vertex absent from its parent. That bag is the unique highest occurrence of the chosen vertex; distinct nonroot bags therefore receive distinct vertices. At most $N + 1$ bags remain. Each contains at most N vertex indices, so canonical serialization has polynomial length with an absolute exponent. Compression itself is within the preceding FPT bound. $\square$

Theorem 7.4 (Tuple FPT and width XP). *Each of REACH, MAX, MIN, ENDPOINTS is FPT in (w_e, q, g_0) on raw input and in (d_e, q, g_0) on supplied input. The task CATEGORYRADIUS is FPT in (w_*, q, g_0) and (d_*, q, g_0) , respectively. Each of these ten task/mode combinations is also in XP for its width parameter alone.*

Proof. Validate first and construct the required summary. Handle $s_0 > n$ directly. For a supplied fixed-radius input, apply Theorem 7.1 and Lemma 7.2; comparison of the maximum with K decides REACH. For raw input first use Lemma 7.3. Its generated width is at most $4w_e + 4$ and its encoded output length is polynomial in B_0 with a fixed exponent. Substitution in (7.5) yields one computable parameter factor times a fixed polynomial in B_0 . Empty summaries are included.

For CATEGORYRADIUS, compute the baseline score and category, then all canonical representatives. At each representative compute both extrema using its own raw or supplied decomposition. Mark the bad cells by Lemma 5.2; take the least bad left endpoint, or infinity, by Theorem 5.3. The singleton at that endpoint determines the boundary bit. There are only $O((B_0 + 1)^2)$ cells and all constructed rational data have polynomial length. Their widths are bounded by exactly w_* or d_* . Replacing the parameter factor by its finite coordinatewise maximum below the given tuple makes it nondecreasing, so summing these polynomially many calls preserves FPT. No common decomposition across radii is assumed.

For XP, in the nontrivial branch $q \leq O(B)$ and $g_0 = s_0 \leq n \leq O(B)$. Equation (7.4) is consequently bounded by $F(d)(B + 1)^{3(d+1)}$ for a computable F . Equation (7.5) then gives a width-dependent polynomial exponent. Substitute $d \leq 4w + 4$ for raw inputs and include the polynomial number of category cells. This is a uniform XP bound. The polynomial invalid-input checks are covered by enlarging the parameter function at zero. $\square$

8. HARDNESS AND THE COST OF EXPLICIT IMAGES

8.1. Classical NP completeness.

Theorem 8.1 (Fixed-weight hardness with one breakpoint). *For every fixed rational pair $a > b > 0$ and every fixed integer $s \geq 2$, REACH is NP-complete with $q = 1$, $s_0 = s$, $(\lambda_0, \lambda_1) = (a, b)$, $\tau = 0$, breakpoint 4, reference radius 1, and zero-area uncertain vertices. Exact MAX and ENDPOINTS are NP-hard. These statements hold in both raw and supplied modes, without a bounded-width claim. In particular $(q, g_0) = (1, 2)$ already gives para-NP-hardness.*

Proof. Membership follows from Corollary 6.3. Reduce INDEPENDENT SET, equivalently the complement-graph version of CLIQUE [10]. We may assume that the simple loopless source graph J

has no isolated vertices: delete them and subtract their number from the requested independent-set size, deciding trivial cases first. Write $n = |V(J)|$, $m = |E(J)|$, $D = a - b$, and

$$B = \left\lfloor \frac{Dn}{b} \right\rfloor + 1, \quad bB > Dn.$$

For each v , create a forced bin-zero center c_v of area 1 and $s - 1$ private forced bin-zero leaves of area 0. Add a zero-area optional vertex d_v adjacent to c_v . For each edge $f = vw$, add a forced bin-one vertex p_f of area B , adjacent only to d_v, d_w . Baselines 2 for centers and leaves, -1 for optional vertices, and 6 for edge vertices realize these domains at the stipulated radius and thresholds.

Let $X = \{v : d_v \text{ is active}\}$ and let $\text{cov}(X)$ count source edges incident to X . An uncovered p_f is a singleton and fails the gate. A covered p_f joins a padded center and qualifies. Since J has no isolated vertices, every selected center joins a bin-one component; an unselected center retains its separate qualifying bin-zero cluster. Thus the score is exactly

$$F(X) = bB \text{cov}(X) + b|X| + a(n - |X|).$$

Every noncover scores at most $bB(m - 1) + an < bBm + bn$, the score of $X = V(J)$. Hence every maximizer is a vertex cover, and among covers it minimizes $|X|$. The identity between minimum vertex-cover size and $n - \alpha(J)$ gives

$$\text{MAX} = b(Bm + n) + D\alpha(J).$$

The threshold $K = b(Bm + n) + Dk$ therefore decides whether $\alpha(J) \geq k$; exact maximum or endpoints recover $\alpha(J)$ by rational arithmetic. For fixed s, a, b , the graph has $O(sn + m)$ vertices and polynomial encoding size; $B = O(n)$. All uncertain vertices have area zero.

To totalize the reduction, replace malformed or already decided source instances by the same construction on a single-edge graph, with independent-set target 1 for YES or 2 for NO. This preserves every fixed restriction. In supplied mode give a one-bag decomposition of the resulting summary. Finally the variable-gate family is hard already at $s = 2$, requiring no construction of size proportional to a binary input gate. $\square$

Figure 2 illustrates the fixed-weight construction on a single source edge.

8.2. A conditional obstruction for width and breakpoint count. The following result retains the cardinality gate on actual source vertices. Freely assigning large binary masses to abstract summary nodes would not establish it.

Theorem 8.2. *If any one of the five tasks REACH, MAX, MIN, ENDPOINTS, CATEGORYRADIUS, in either raw or supplied mode, has an FPT algorithm parameterized jointly by its specified width and q , then $\text{FPT} = \text{W}[1]$. For REACH the reduction is a parameterized many-one reduction; for the four function tasks it uses one exact-output call and explicit postprocessing.*

Proof. Unary Bin Packing is $\text{W}[1]$ -hard parameterized by the input number k_{in} of bins [9, Theorem 2]. Items have unary-encoded positive sizes; capacity C and bin count k_{in} are integers and need not be unary. We first give a total polynomial normalization to exact-fill instances.

For the normalization only, write $k = k_{\text{in}}$. Let item sizes be $a_1, \dots, a_m$ and total W . Parse with length guards. Invalid input, an item larger than C , or $W > kC$ is a known NO case; empty input is a known YES case. After checking positive k, C and that every item fits, the cases $k \geq m$ and $C \geq W$ are also YES. Otherwise $k < m$ and $C < W$, so both are bounded by the unary input length. Append $kC - W$ unit items. The new list has polynomial length and total kC . It is packable exactly when the original was: fill every bin's unused capacity with unit items in the forward direction and delete filler items in the reverse direction. Known cases may instead be replaced by the fixed exact-fill YES instance $(k, C; a) = (2, 1; 1, 1)$ or NO instance $(2, 3; 2, 2, 2)$. Henceforth k denotes the normalized bin count, so $k \leq \max(k_{\text{in}}, 2)$ on every branch. We have $a_i > 0$ and $\sum_i a_i = kC$, and write m for the padded item count.

Source graph: one edge v_1v_2 ; $s_0 = 2$, $(\lambda_0, \lambda_1) = (2, 1)$, $B = 3$
 $a(c_1) = a(c_2) = 1$, $a(p) = 3$, $a(\ell_1) = a(\ell_2) = a(d_1) = a(d_2) = 0$

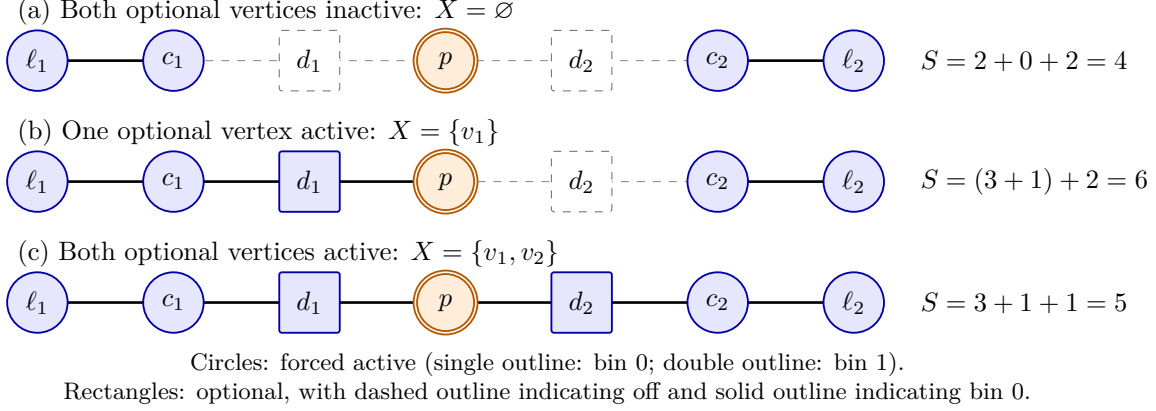


FIGURE 2. The fixed-weight hardness gadget for a single source edge. This artificial graph uses zero-area auxiliary vertices: area and cardinality are distinct. Each pair $\{\ell_i, c_i\}$ meets the cardinality gate, whereas p alone does not. Activating d_1 makes p contribute, but lowers the weight on the area at c_1 from 2 to 1. Activating d_2 as well only incurs another such loss. Thus an optimum activates a minimum vertex cover; its complement, represented by inactive optional vertices, is a maximum independent set. The construction is realizable with $\tau = 0$, $b_1 = 4$, $\varepsilon = 1$, baseline intensities 2 for forced bin-0 vertices, 6 for p , and -1 for optional vertices.

Put $L = m + 1$. Create stable bin terminals $z_1, \dots, z_k$. For item i , create a path P_i of La_i vertices with endpoint r_i . For each pair (i, j) create a connector u_{ij} adjacent exactly to r_i and z_j . All item vertices and connectors have area zero. Set

$$\begin{aligned} \tau = 0, \quad e = 1, \quad q = k, \quad b_j = 4j, \quad \lambda_j = k + 1 - j \quad (0 \leq j \leq k), \\ x(P_i) = 2, \quad x(z_j) = 4j + 2, \quad a(z_j) = 1/\lambda_j, \quad x(u_{ij}) = -\frac{1}{2}, \\ s_0 = 1 + LC, \quad K = k. \end{aligned} \tag{8.1}$$

At $e = 1$, each item path is a stable bin-zero component of mass La_i . Each terminal is stable in bin j , and each connector permits exactly inactivity or active bin zero. The source has $k + L \sum_i a_i + km$ vertices, polynomial in the unary input length. Every mass is realized by that many actual vertices.

Consider any active component containing terminal set $J \neq \emptyset$. Its maximum bin is $j^* = \max J$. If it meets the gate, its score is

$$\sum_{j \in J} \frac{\lambda_{j^*}}{\lambda_j} \leq |J|,$$

with equality exactly when $|J| = 1$, because the positive λ_j strictly decrease. If it fails the gate, its score is zero, strictly below $|J|$. Components without terminals have zero area. Summing over disjoint terminal sets proves that the total score is at most k , with equality exactly when each terminal belongs to a distinct qualifying component.

If equality holds, an item path can connect to at most one terminal; otherwise two terminal components would merge. Assign it to that terminal if connected. A component containing z_j has cardinality

$$1 + L \sum_{i \text{ assigned to } j} a_i + c_j,$$

where c_j is its number of active connectors. There is at most one connector for each item-terminal pair, so $c_j \leq m < L$. Meeting the gate (8.1) forces the integer item sum to be at least C . Summing over k disjoint item groups and using total kC forces equality at every bin and assignment of every item. This is a perfect packing. Conversely, a perfect packing activates just its item-to-bin connectors; the k terminal components remain separate, meet the gate, and each score one. Thus reachability of k is equivalent to packing.

We now bound every relevant width and construct the supplied records. The source graph has a decomposition with root bag $\{z_1, \dots, z_k\}$. For each item, attach a bag containing all terminals and r_i . Attach to it the bags $\{r_i, z_j, u_{ij}\}$ for each j , and a chain of two-vertex bags for the remaining path edges. Every edge is covered and every vertex has connected occurrences. The width is at most $\max(k, 2)$.

At an arbitrary radius, the interaction graph is obtained by deleting stable-inactive vertices and contracting each stable-active connected component. Map every original bag through this deletion and contraction, removing duplicates and retaining empty bags. Its size cannot increase. The occurrence set of a contracted node is the union of the occurrence subtrees of its source vertices. This union is connected: consecutive vertices along a spanning tree of the contracted component have a common edge-containing bag. Summary edges are covered by their witnessing source edges. Hence the mapped bags are a valid decomposition of every H_e , with width at most $\max(k, 2)$. In particular, $w_e, d_e, w_*, d_* \leq \max(k, 2) \leq \max(k_{\text{in}}, 2)$, while $q = k \leq \max(k_{\text{in}}, 2)$. This proves both the fixed-radius and the whole-radius bounds. For supplied inputs serialize the appropriate mapped decomposition, or generate all canonical representatives and serialize their mapped family. The number and lengths of records are polynomial.

An FPT REACH algorithm would now solve Unary Bin Packing in parameter k . An exact MAX or ENDPOINTS call decides whether the maximum is k . For MIN, negate every density weight; the score is negated while domains and widths are unchanged, so the test is whether the minimum is at most $-k$.

For CATEGORYRADIUS, keep the original weights and use one category cutpoint k . At the baseline every connector is inactive. Terminals have mass one below s_0 , and all other components have zero area, so the baseline score is zero. A bad perturbation at radius one exists exactly when the packing exists. Perturbation boxes are nested, hence Bad is upward closed. Given the exact radius answer, radius one is bad precisely when

$$d < 1 \quad \text{or} \quad (d = 1 \text{ and } v_d = 0); \quad (8.2)$$

infinity gives NO. Indeed, if $d < 1$, some bad rational radius is below one, since otherwise one would be a lower bound for Bad; nestedness then gives badness at one. If $d = 1$, the boundary bit decides exactly. The converse follows from the definition of infimum. The every-radius decomposition bound controls w_* and d_* in this reduction.

Finally, suppose one of the ten algorithms runs in $f(w, q)(B+1)^c$, where w is that task's specified width. Put $h(k_{\text{in}}) = \max(k_{\text{in}}, 2)$ and replace f by the finite bound

$$F(k_{\text{in}}) = \max\{f(u, v) : 0 \leq u, v \leq h(k_{\text{in}})\}.$$

This is computable even when f is not monotone. The reduction has polynomial output length, and all comparisons and exact-output postprocessing have polynomial bit cost. Any of the ten algorithms would therefore put Unary Bin Packing in FPT, implying $\text{FPT} = \text{W}[1]$. $\square$

Theorem 8.2 rules out omitting g_0 from the positive tuple (width, q, g_0) unless $\text{FPT} = \text{W}[1]$. It does not settle the pair (width, g_0) with unrestricted q , or prove hardness at constant width. Indeed, the cardinality threshold in the construction is not bounded by the packing parameter: exact-fill items $(t, t-1, 1)$ with $k = 2$ and $C = t \geq 2$ give $L = 4$ and $g_0 = 1 + 4t$. This illustrates the unbounded threshold, rather than hardness on a fixed- k slice.

8.3. Positive-weight hardness with two breakpoints.

Theorem 8.3 (Two-breakpoint positive-weight hardness). *REACH is NP-complete even with $q = 2$, $s_0 = g_0 = 1$, $e = 1$, $\tau = 0$, breakpoints $(4, 8)$, nonnegative integer areas, and strictly positive strictly decreasing rational weights. Exact MAX is NP-hard under a one-call exact-output reduction. The three numerical weights are input-dependent. These statements hold in both modes, without a bounded-width claim.*

Proof. Use unweighted three-terminal Multiway Cut, NP-complete by Dahlhaus et al. [6, Theorem 3]. The source may be a loopless multigraph: the distinct switch vertices introduced below make the constructed source graph simple even when source edges are parallel.

Let $H = (W, F)$ have distinct terminals t_0, t_1, t_2 , $N = |F| \geq 1$, and minimum separating cut size c . Define

$$\delta = \frac{1}{4N}, \quad M = 8N^2 + 1, \quad (\lambda_0, \lambda_1, \lambda_2) = (1 + 2\delta, 1 + \delta, 1), \quad B_{\text{term}} = M(\lambda_0 + \lambda_1 + \lambda_2).$$

Subdivide every edge $f = uv$ by a switch z_f . Give z_f area 1, baseline -1 ; all original nonterminals have area zero and baseline 1; the terminal t_i has area M and baseline $4i + 1$. With the fixed data in the theorem, switch intervals are $[-2, 0]$ and terminal intervals are $[4i, 4i + 2]$. Thus switches independently choose off or bin zero, all originals remain active, and terminal bins are fixed. Chosen switches can take intensity 0, unchosen switches -2 . This is an actual common-radius source construction.

For active switch set $A \subseteq F$, connectivity among originals is exactly that in (W, A) . Let $\mathcal{P}(A)$ partition $\{0, 1, 2\}$ by terminal connectivity. Let $j(f)$ be the largest terminal index in the component of switch f , or zero for a terminal-free component. Since the gate is one,

$$S(A) = M \sum_{T \in \mathcal{P}(A)} |T| \lambda_{\max T} + \sum_{f \in A} \lambda_{j(f)}. \quad (8.3)$$

The formula includes untouched singleton terminals and terminal-free components.

Put $L(\mathcal{P}) = \sum_{T \in \mathcal{P}} \sum_{i \in T} (\max T - i)$. The terminal contribution is $B_{\text{term}} - M\delta L(\mathcal{P})$. Its loss is zero for the fully separated partition and at least $M\delta$ for any merger. Explicitly the five terminal partitions give $L = 0, 1, 2, 1, 3$. Switch contributions are at most

$$N(1 + 2\delta) = N + \frac{1}{2} < M\delta = 2N + \frac{1}{4N}.$$

Every merged assignment therefore scores below B_{term} , whereas switching all edges off scores B_{term} . Every maximizer separates the terminals.

For a separating assignment, $F \setminus A$ is a cut, hence $|A| \leq N - c$. Conversely a minimum cut leaves $N - c$ switches available. Each switch contributes between 1 and $1 + 2\delta$, proving

$$B_{\text{term}} + N - c \leq S_e^+ \leq B_{\text{term}} + N - c + \frac{1}{2}, \quad c = N - \lfloor S_e^+ - B_{\text{term}} \rfloor. \quad (8.4)$$

For a source bound $0 \leq k \leq N$, set

$$K = B_{\text{term}} + N - k = \frac{M(12N + 3) + 4N(N - k)}{4N},$$

reduced to canonical form. If $c \leq k$, the threshold is attained or exceeded; if $c > k$, then $S_e^+ \leq K - \frac{1}{2}$. Thus the reduction also proves threshold-decision hardness. Membership in NP follows from the legal state-vector certificate and exact polynomial-bit score evaluation.

The graph has $|W| + N$ vertices and $2N$ edges; M , the weights, B_{term} , and K have $O(\log(N + 1))$ bits for $0 \leq k \leq N$. Validate the source first: negative k and malformed encodings are NO, $k \geq N$ is YES, so $N = 0$ is handled before defining δ . Fixed YES/NO target instances can use a single zero-area vertex, the advertised constant thresholds, weights $(3/2, 5/4, 1)$, and threshold 0 or 1. In supplied mode a single bag containing the entire summary is a valid polynomial-size record. No bounded-width consequence is claimed. $\square$

For $s_0 = 1$, this establishes a distinction between one and two breakpoints for positive-weight MAX when weights are allowed to vary as part of the input: $q = 0$ is a total-active-area calculation, and the one-breakpoint case in Section 10 is polynomial. It does not prove hardness for any fixed numerical three-weight vector. Replacing δ by a fixed positive constant would lose the half-unit decoding bound as N grows.

8.4. The explicit-image output barrier.

Theorem 8.4. *Explicit sorted IMAGE cannot be computed in input-polynomial total time even at the fixed parameter tuple $(w_e, q, g_0) = (0, 0, 1)$. The same statement holds for supplied width zero.*

Proof. For $n \geq 1$, take an edgeless graph, $q = 0$, $\lambda_0 = 1$, $s_0 = 1$, $\tau = 0$, $x_i = -1$, $e = 1$, and $a_i = 2^i$ for $0 \leq i < n$. Each vertex independently permits inactivity or active bin zero, and every active singleton qualifies. Binary expansion gives the exact image

$$\mathcal{R}_e = \{0, 1, \dots, 2^n - 1\}.$$

The summary is edgeless. The input length is $\Theta(n^2)$: area numerators have successively increasing binary lengths, and other fields contribute only $O(n \log(n+1))$. A path of singleton bags supplies width zero with the same asymptotic input length. The exact rational output contains 2^n records and has $\Theta(n2^n)$ bits, since at least half its numerators have n binary digits. Output emission alone exceeds every polynomial in input length. This is exponential in n , not a claim of $2^{\Omega(B)}$ growth. $\square$

The obstruction concerns explicit listing. It does not preclude polynomial time for the short-output tasks on this family, nor does it settle output-sensitive enumeration or delay bounds.

9. THE NONNEGATIVE MONOTONE WEIGHT SUBFAMILY

The general model permits signed and nonmonotone weights. The following additional result imposes an explicit restriction only within this section; none of the preceding theorem hypotheses is changed.

Proposition 9.1 (Monotonicity and corner extrema). *Suppose, in addition to the model assumptions of Section 2, that*

$$0 \leq \lambda_0 \leq \lambda_1 \leq \dots \leq \lambda_q. \quad (9.1)$$

Then for all $y, z \in \mathbb{Q}^V$ with $y_v \leq z_v$ at every vertex,

$$S_G(y) \leq S_G(z). \quad (9.2)$$

Consequently, for every rational baseline x and radius $e \geq 0$,

$$S_e^- = S_G(x - e\mathbf{1}), \quad S_e^+ = S_G(x + e\mathbf{1}), \quad (9.3)$$

where $(x \pm e\mathbf{1})_v = x_v \pm e$. On this restricted subfamily, REACH, MAX, MIN, ENDPOINTS, and CATEGORYRADIUS admit deterministic polynomial-bit-time exact algorithms in both raw and supplied-decomposition modes, including the separate finite-boundary invariance bit. The polynomial bound uses the complete input length, with no fixed-width, fixed-bin-count, or bounded-gate hypothesis.

Proof. Since $y \leq z$, the active set at y is contained in the active set at z . Every active component C at y therefore lies in a unique active component D at z . If $|C| \geq s_0$, then $|D| \geq s_0$. Moreover,

$$\max_{v \in C} y_v \leq \max_{v \in D} z_v, \quad \text{bin}\left(\max_{v \in C} y_v\right) \leq \text{bin}\left(\max_{v \in D} z_v\right).$$

Write these bin indices as J_C and J_D . For a fixed D , let $\mathcal{F}_D$ be the family of qualifying y -components contained in it. Their supports are disjoint. Nonnegative areas and (9.1) imply

$$\sum_{C \in \mathcal{F}_D} a(C) \lambda_{J_C} \leq \lambda_{J_D} \sum_{C \in \mathcal{F}_D} a(C) \leq \lambda_{J_D} a(D), \quad (9.4)$$

where $a(C) = \sum_{v \in C} a_v$. Every nonempty $\mathcal{F}_D$ has a qualifying D . Sum (9.4) over those components. Any additional qualifying component at z contributes a nonnegative amount, so including it preserves the inequality. This proves (9.2); it also covers zero areas, newly qualifying components, and mergers.

Every legal perturbation satisfies $x - e\mathbf{1} \leq y \leq x + e\mathbf{1}$, and both rational corners belong to $P_e(x)$. Applying (9.2) on both sides proves (9.3) with attainment.

For complexity, first perform the complete task validation from Theorem 6.2 and check (9.1) by exact comparisons. In supplied mode this includes every required decomposition record and its original graph-specific validity conditions, even though the optimization step does not need a decomposition. At the encoded fixed radius, form the two corner vectors, compute their states, and evaluate their scores. Lemma 6.1 and Theorem 6.2 give polynomial bit lengths and running time. The two values answer MAX, MIN, and ENDPPOINTS; comparison of the upper one with K answers REACH.

For CATEGORYRADIUS, compute the baseline category and all canonical critical cells from Section 5. At each representative evaluate the two corners and apply Lemma 5.2. Theorem 5.3 then returns infinity or the least bad-cell left endpoint; the two corner scores at its singleton cell give the exact boundary bit. There are polynomially many cells, each with polynomially encoded representatives, so the complete computation and output are polynomial in the input length. This argument does not assume that the infimum is attained or that the image fills its hull.

The algorithms retain the original rejection/error behavior on invalid strings. To regard them as total exact algorithms for the original unrestricted task language, on a valid input that fails (9.1) use the general finite-state procedure instead. Correctness is then total, while the polynomial-time claim here is exactly for the restricted subfamily and invalid inputs. In particular, neither an extra promise about supplied decompositions nor a change to the original task encoding is required. $\square$

Remark 9.2. Nonnegativity cannot be omitted: Proposition 10.4 gives hardness for fixed negative increasing weights. It is used in the last inequality of (9.4) and when adding newly qualifying components; it is part of the hypothesis, not a consequence of nondecreasing weights alone. This easy subfamily still permits arbitrary finite graphs, rational precision, cardinality gates, and density-breakpoint counts. The explicit-image family in Theorem 8.4 has $q = 0$ and $\lambda_0 = 1$, so its output-size obstruction already satisfies (9.1).

9.1. Common offsets and joint threshold translation. Write $S_{\tau,b}$ when the intensity thresholds need emphasis, and define

$$D_e(x) = \{x + c\mathbf{1} : c \in \mathbb{Q}, |c| \leq e\} \subseteq P_e(x).$$

Corollary 9.3 (Common-offset equivalence for short outputs). *For arbitrary rational weights and every $c \in \mathbb{Q}$,*

$$S_{\tau,b}(x + c\mathbf{1}) = S_{\tau-c,b-c\mathbf{1}}(x). \quad (9.5)$$

Under (9.1), the two perturbation families have the same attained extrema,

$$\begin{aligned} \min S(P_e) &= \min S(D_e) = S(x - e\mathbf{1}) = S_{\tau+e,b+e\mathbf{1}}(x), \\ \max S(P_e) &= \max S(D_e) = S(x + e\mathbf{1}) = S_{\tau-e,b-e\mathbf{1}}(x). \end{aligned}$$

They have the same baseline-category invariance predicate for every rational radius, the same category radius, and the same finite-boundary bit. Consequently their answers to all five short-output tasks agree.

Proof. The equivalences $x_v + c \geq \tau \iff x_v \geq \tau - c$ and $b_j \leq t + c \iff b_j - c \leq t$ preserve active components and selected bins, including threshold equalities. This proves (9.5). Under (9.1), Proposition 9.1 gives the extrema at the two corners, which belong to both families. A nondecreasing category map is constant on an attained score image exactly when its two extremal values have that category. Invariance therefore agrees pointwise for every rational radius. The bad-radius sets

are equal, so are their extended-real infima, including infinity, and their separate finite-boundary tests. Finite radii are rational by Theorem 5.3. $\square$

Example 9.4 (Different full images). For two isolated unit-area vertices, take $x = (0, 0)$, $e = 1$, $\tau = 0$, $q = 0$, and $s_0 = \lambda_0 = 1$. The independent box has score image $\{0, 1, 2\}$, whereas common offsets have image $\{0, 2\}$. Category cutpoints 1, 2 give category images $\{0, 1, 2\}$ and $\{0, 2\}$ as well. Equal extrema and equal invariance do not imply equal full images.

All density breakpoints must be translated in (9.5). For one unit-area vertex with $x = 3/2$, $c = 1/2$, $\tau = 0$, $b_1 = 2$, weights $(1, 2)$ and gate one, the shifted score is 2. Translating only τ to $-1/2$ leaves score 1; also translating b_1 to $3/2$ restores equality. Thus changing only a detection threshold is a different operation.

10. ONE BREAKPOINT: THE ROLE OF THE CARDINALITY THRESHOLD

Throughout this section $q = 1$ and $b = b_1$. The graph is part of the input and is fixed during perturbation; areas are nonnegative. Put

$$W_+ = \{(\lambda_0, \lambda_1) : \lambda_0 > \lambda_1 > 0\}, \quad W_- = \{(\lambda_0, \lambda_1) : \lambda_0 < \lambda_1 < 0\}.$$

Theorem 10.1 (Fixed-pair classification for all cardinality thresholds). *For every fixed rational weight pair, the following table gives NP-hard regions; outside the indicated region each task has a polynomial-bit-time algorithm. The second row applies to every fixed $s_0 \geq 2$, and also to the family where s_0 is part of the input. All statements hold in raw and supplied modes.*

Gate	MAX, REACH	MIN	ENDPOINTS, CATEGORYRADIUS
$s_0 = 1$	W_-	W_+	$W_+ \cup W_-$
Fixed $s_0 \geq 2$; input s_0	$W_+ \cup W_-$	$W_+ \cup W_-$	$W_+ \cup W_-$

Hardness for a function means a polynomial construction followed by one exact-output call and polynomial-time postprocessing. For REACH it strengthens to NP-completeness. The algorithms are uniform for input rational pairs in their tractable regions and, in the second row, for a binary-encoded input gate.

The axes and diagonal belong to the polynomial regions. An exclusive interpretation of “hard versus polynomial” requires $P \neq NP$. For $s_0 = 1$, the tractable region for MAX is precisely the union of $\lambda_0 \geq \lambda_1$ and $\lambda_1 \geq 0$: use the closure below in the first region, the upper corner when $0 \leq \lambda_0 \leq \lambda_1$, and the mixed-sign construction when $\lambda_0 \leq 0 \leq \lambda_1$. For arbitrary gates, the two monotone same-sign regions and the two mixed-sign regions are polynomial, as proved below. Theorem 8.1 and the padded Set Cover reductions prove the remaining hardness.

The identity $\text{MIN}_\lambda = -\text{MAX}_{-\lambda}$ gives the minimum and endpoint statements. Both extrema evaluated on the finitely many exact critical cells give the radius algorithm by Theorem 5.3; a separate reduction proves radius hardness. All procedures validate the full input, including supplied decompositions, and obtain rational witnesses through Lemma 3.2. Finite-state enumeration remains an exact fallback outside a stated polynomial region.

10.1. Legal domains and a uniform two-layer closure. Let

$$\begin{aligned} P &= \{v : U_v \geq \tau\}, & F &= \{v \in P : L_v \geq \tau\}, \\ B_1 &= \{v \in P : U_v \geq \max(\tau, b)\}, & Z_v &\iff \tau < b, U_v \geq \tau, L_v < b. \end{aligned}$$

Here P consists of possibly active vertices, F of forced-active vertices, B_1 of bin-one-capable vertices, and Z_v says active bin zero is allowed. Work in $H = G[P]$. A stable-active vertex can belong to B_1 even if its lower endpoint is in bin zero.

Proposition 10.2 (Uniform closure formula). *Assume $s_0 = 1$ and $\lambda_0 \geq \lambda_1$. Introduce binary variables X_v, Y_v for $v \in P$, with constraints*

$$X_v \leq Y_v \quad (v \in P), \quad (10.1)$$

$$X_u \leq Y_v, \quad X_v \leq Y_u \quad (uv \in E(H)), \quad (10.2)$$

$$Y_v \leq X_v \quad (v \in F), \quad (10.3)$$

$$Y_v = 1 \quad (v \in P \text{ with } Z_v \text{ false}). \quad (10.4)$$

Then, with $K_0 = \lambda_0 a(P)$,

$$S_e^+ = K_0 - \min \left\{ C(X, Y) = \sum_{v \in P} a_v(-\lambda_1 X_v + \lambda_0 Y_v) \right\}. \quad (10.5)$$

The minimum is a polynomial-bit-time minimum-cut problem, and a minimizing solution can be repaired constructively into an attaining rational perturbation.

Proof. The intended labels $(X, Y) = (0, 0), (0, 1), (1, 1)$ mean respectively a vertex in a bin-zero component, an off vertex, and a vertex in a bin-one component. X labels the *component type*, not the vertex's individual bin. In particular, $X_v = 1$ need not imply $v \in B_1$.

Every actual state satisfies the constraints: (10.2) forbids an edge between distinct active component types, (10.3) forbids deactivating a forced vertex, and (10.4) forbids an unavailable zero-bin assignment. Its actual score equals $K_0 - C$. This proves the right side of (10.5) is an upper bound.

Conversely take any feasible pair, and put $R = \{v : X_v = 1\}$. A connected component D of $H[R]$ is *witnessless* if $D \cap B_1 = \emptyset$. Each vertex of D allows bin zero, since it can be active but cannot attain bin one. Change both its variables to zero. Constraints remain satisfied: an external selected X_u adjacent to D would belong to the same component of $H[R]$, so no such antecedent exists; forced-activity inequalities still hold; and no vertex of D is subject to (10.4). The cost changes by

$$-(\lambda_0 - \lambda_1)a(D) \leq 0.$$

Remove all such components. In every surviving component of $H[R]$, choose a bin-one-capable vertex as witness, and give all other vertices any allowed active bin. Assign $(0, 0)$ vertices bin zero and $(0, 1)$ vertices off. There is no active edge joining R to the zero-bin region, so this realizes exactly $K_0 - C$ for the repaired pair. Starting with a minimum, the cost cannot strictly improve; thus it is unchanged and equality follows. This repair matters when equal weights or zero areas permit witnessless minimizers.

For completeness, reduce the closure to a cut explicitly [11]. Make one node for each variable. An implication $A \leq D$ becomes a directed hard arc $A \rightarrow D$; a forced-one variable gets a hard source arc. A node with cost $c \geq 0$ gets a sink arc of capacity c ; one with $c < 0$ gets a source arc of capacity $-c$. Source-side nodes are selected. Multiply all rational costs by the product D_0 of their denominators, and set each hard capacity to

$$M_0 = 1 + \sum_z |D_0 c_z|.$$

All variables selected is feasible, so a minimum cut never crosses a hard arc. If its value is κ , then

$$\min C = \frac{\kappa + \sum_{c_z < 0} D_0 c_z}{D_0}.$$

The product D_0 has polynomial bit length because denominator bit lengths add. The cut graph has $O(n)$ nodes and $O(n + |E|)$ arcs. A shortest-augmenting-path algorithm uses polynomially many arithmetic operations independently of capacity magnitudes [7]; all capacities have polynomial bit length. Reconstruction and rational output reduction also have polynomial bit cost. If P is empty, the score is zero directly. $\square$

10.2. A boundary formulation with ordered thresholds. For this simpler boundary formulation assume $s_0 = 1$ and $\lambda_0 \geq \max(\lambda_1, 0)$. When $\tau < b$, let $M_1 = \{v \in P : Z_v \text{ is false}\}$; these are exactly the forced bin-one vertices. For $R \subseteq P$ with $M_1 \subseteq R$ and $\partial R \cap F = \emptyset$, where $\partial R = N_H(R) \setminus R$, the simpler formula is

$$S_e^+ = \lambda_0 a(P) - \min_R ((\lambda_0 - \lambda_1) a(R) + \lambda_0 a(\partial R)). \quad (10.6)$$

Initially require every component of $H[R]$ to meet B_1 : activate R with witnesses, deactivate ∂R , and activate every remaining vertex in bin zero. Conversely any state can activate additional off vertices outside $R \cup \partial R$ in bin zero without reducing the score. The witness requirement can then be dropped in minimization: removing a witnessless component removes no mandatory vertex and introduces no new boundary; both coefficients in the displayed cost are nonnegative. An optimal relaxed set repairs to an equally good witnessed set.

Example 10.3 (A source-realizable failure at $\tau = b$). Take one isolated area-one vertex, $\tau = b = 0$, $e = 1$, $x = -1$, and $(\lambda_0, \lambda_1) = (1, -1)$. Its interval $[-2, 0]$ allows exactly off or bin one. Its score image is $\{0, -1\}$, so $\text{MAX} = 0$. The unrestricted boundary expression gives 1 by $R = \emptyset$. If residual-bin-zero feasibility is added, only $R = \{v\}$ remains and gives -1 , still missing the optimal off state. An optional bin-one-only vertex may need to be off without being a boundary vertex. The two-layer formula handles this with $(X, Y) = (0, 1)$.

A separate branch is also possible: when $\tau \geq b$, every active component has weight λ_1 . Thus $\text{MAX} = \lambda_1 a(P)$ if $\lambda_1 \geq 0$, and $\text{MAX} = \lambda_1 a(F)$ if $\lambda_1 < 0$.

10.3. Polynomial regions for arbitrary cardinality thresholds. For $0 \leq \lambda_0 \leq \lambda_1$, Proposition 9.1 gives the upper-corner maximum for every gate. For $\lambda_1 \leq \lambda_0 \leq 0$, negate the weights and use its lower-corner minimum. It remains to prove both mixed-sign regions. All component sizes in this argument count original source vertices.

Suppose first that $\lambda_0 \leq 0 \leq \lambda_1$. For every component D of $H = G[P]$ containing a vertex of B_1 , activate all of D with a bin-one witness. If $|D| < s_0$, every attainable contribution inside D is zero. Otherwise this attains $\lambda_1 a(D)$: in any assignment qualifying high components contribute at most this amount and qualifying low components contribute nonpositive amounts. If $D \cap B_1 = \emptyset$, activate exactly $F \cap D$. All active vertices there have bin zero. Adding active vertices can only increase the total area in qualifying components, since every previously qualifying component embeds in a new qualifying component. With coefficient $\lambda_0 \leq 0$, the forced-active set maximizes the score.

Now suppose $\lambda_1 \leq 0 \leq \lambda_0$. If $\tau \geq b$, bin zero is unavailable and the same qualifying-area argument shows that activating exactly F is optimal. If $\tau < b$, let

$$M_1 = \{v \in F : Z_v \text{ is false}\}, \quad Q = \bigcup \{C \in \text{cc}(G[F]) : C \cap M_1 \neq \emptyset\},$$

$$B = N_H(Q) \setminus Q, \quad R = P \setminus (Q \cup B).$$

Every optional possibly active vertex allows bin zero, because its interval crosses $\tau < b$. Also $B \cap F = \emptyset$, since any forced neighbor of Q belongs to the same forced component. Keep Q active, deactivate B , and activate all of R in bin zero. This is feasible: any forced vertex lacking bin zero belongs to $M_1 \subseteq Q$. If $Q_1, \dots, Q_j$ are the forced components comprising Q , the score is

$$\lambda_1 \sum_{i: |Q_i| \geq s_0} a(Q_i) + \lambda_0 \sum_{C \in \text{cc}(G[R]), |C| \geq s_0} a(C).$$

In any assignment every qualifying Q_i remains inside a qualifying high component, so the nonpositive high contribution is at most the first term. Any low component must avoid both Q and its neighbors B ; its qualifying area is bounded by the qualifying area of $G[R]$, giving the second upper bound. Our assignment attains both simultaneously. This proves optimality, including zero weights, zero areas, and gates larger than the graph.

These constructions require only connected components, neighbor sets, rational sums, and comparisons with counts at most n . A binary-encoded gate creates no pseudopolynomial loop. Zero-area vertices are retained throughout because they can connect components, meet the gate, or witness a high bin.

10.4. Vertex-Steiner hardness. The precise source problem needed is the minimum number of *nonrequired vertices* in a connected vertex set containing a root r and terminals T . This version is NP-hard by an explicit Set Cover reduction: for universe $U = \bigcup_{i=1}^N S_i$, make a root, one optional vertex w_i per set, and one required terminal t_u per element; add rw_i and w_it_u exactly when $u \in S_i$. A connected set spanning all terminals exists with at most k optional vertices exactly when there is a cover by at most k sets. Set Cover is one of Karp's complete problems [10, Main Theorem, item 6]. Malformed or trivial instances can be decided first and replaced by fixed instances: the one-element, one-set instance with budget 1 is YES and the same instance with budget 0 is NO. Thus restrict to nonempty covered universes and $0 \leq k \leq N$; the resulting graph is connected. A proposed connected vertex set is checked in polynomial time, so the corresponding vertex-count decision problem is NP-complete.

Proposition 10.4 (Hardness for every negative increasing fixed pair). *Fix rationals $\alpha < \beta < 0$. For $q = 1$ and every fixed $s_0 = s \geq 1$, exact MAX is NP-hard at $(\lambda_0, \lambda_1) = (\alpha, \beta)$, even with fixed intensity thresholds and reference radius. So is REACH with an input rational threshold.*

Proof. Use the root-set-element graph just constructed. Let $t = |T| \geq 1$, let N count set vertices, and let κ be the minimum cover size. Put $\Delta = \beta - \alpha > 0$ and

$$M = 1 + \left\lceil \frac{(-\alpha)(N+1)}{\Delta} \right\rceil, \quad \text{so} \quad \Delta M > (-\beta)N.$$

Attach $s - 1$ private forced bin-zero leaves of area zero and baseline 2 to the root and to every terminal. Use $\tau = 0$, $b = 4$, $e = 1$, root baseline 6 and area 0, terminal baselines 2 and areas M , and optional baselines -1 and areas 1. The intervals $[5, 7]$, $[1, 3]$, and $[-2, 0]$ give precisely a fixed bin-one root, fixed bin-zero terminals, and off/bin-zero optional vertices.

For an arbitrary assignment, let h count uncovered terminals and u count selected set vertices. Every selected set vertex is adjacent to the root. Every uncovered terminal retains its separate size- s padded cluster; the root cluster also qualifies. Hence the exact score is

$$F = \beta(M(t - h) + u) + \alpha Mh = \beta Mt - \Delta Mh + \beta u. \quad (10.7)$$

If $h \geq 1$, then $F \leq \beta Mt - \Delta M < \beta(Mt + N)$, the score obtained by selecting every set. Thus every maximizer covers all terminals and minimizes u , because $\beta < 0$. Therefore

$$\text{MAX} = \beta(Mt + \kappa), \quad \kappa \leq k \iff \text{MAX} \geq \beta(Mt + k). \quad (10.8)$$

Every assignment is realized by optional coordinate 0 when selected and -1 otherwise, with root and terminal baselines retained. For a fixed pair, $M = O(N+1)$ and its bit length is $O(\log(N+1))$. Graph encoding and rational thresholds have polynomial size. Exact output decodes $\kappa = \text{MAX}/\beta - Mt$. The same arithmetic remains polynomial-bit if the strict pair is supplied as input, even when Δ is small. $\square$

Negate all weights and scores in Proposition 10.4. For any fixed positive decreasing pair $a > b > 0$, the same construction gives

$$\text{MIN} = b(Mt + \kappa), \quad \kappa \leq k \iff \text{MIN} \leq b(Mt + k), \quad (10.9)$$

where $M = 1 + \lceil a(N+1)/(a-b) \rceil$. Together with Theorem 8.1 and sign reversal, this proves the extremum hardness statements of Theorem 10.1. A supplied one-bag decomposition gives the same classical hardness in supplied mode without controlling width.

Proposition 10.5 (Category radius for fixed pairs and arbitrary fixed gates). *For $q = 1$ and every fixed $s_0 = s \geq 1$, CATEGORYRADIUS is NP-hard in either strict wedge $\lambda_0 > \lambda_1 > 0$ or $\lambda_0 < \lambda_1 < 0$, for every fixed rational pair in that wedge. It is polynomial-bit-time computable outside their union. In either hard wedge the reduction below has exact answer $(1, 0)$ on YES instances and $(2, 0)$ on NO instances.*

Proof. Use the padded root-set-element graph and baseline above, with the positive pair, the displayed M , and the sole category cutpoint

$$c = b(Mt + k + \tfrac{1}{2}).$$

At the baseline all optional vertices are off and each terminal belongs to its separate padded qualifying cluster. Hence $S(x) = aMt > c$, since

$$(a - b)Mt > a(N + 1)t \geq a(N + 1) > b(k + \tfrac{1}{2}),$$

where $t \geq 1$ and $0 \leq k \leq N$. The baseline category is 1; a category change is exactly a score strictly below c . There can be no unintended upward-category crossing with a single cutpoint.

At radius 1, (10.9) gives $S_1^- = b(Mt + \kappa) < c$ exactly when $\kappa \leq k$. For $0 \leq e < 1$, all state domains equal their baseline singletons, so no radius is bad. For $1 \leq e < 2$, the domains are exactly those at 1. Thus a NO instance has no bad radius below 2. At $e = 2$, every terminal can take intensity 4, hence bin one. Leave all optional vertices off and retain root intensity 6. The padding leaves remain active even at radius 2, and have area zero. This scores $bMt < c$, so 2 is always bad. Equality at the thresholds is legal, giving the asserted boundary bits. All numerical data have polynomial bit length.

For a fixed negative increasing pair $(\alpha, \beta) = (-a, -b)$, use the same graph and areas and the cutpoint $c' = -c$. The baseline score is $-aMt < c'$, so its category is 0 and a change means score at least c' . At radius 1 its maximum is $-b(Mt + \kappa)$, which is at least $-b(Mt + k + 1/2)$ exactly when $\kappa \leq k$. The half-integer gap excludes equality in this comparison. The domains are unchanged on $[0, 1)$ and $[1, 2)$, and at radius 2 the all-high-terminal state gives $-bMt > c'$. Hence the same exact outputs $(1, 0)$ and $(2, 0)$ follow directly with the correct category convention.

Outside the two wedges, Theorem 10.1 supplies polynomial algorithms for both extrema. Evaluate them on the polynomially many singleton/open critical-radius cells and test the boundary separately, by Theorem 5.3. This proves the remaining tractability statement. $\square$

When $s_0 = 1$, positive decreasing weights give polynomial MAX but hard MIN, ENDPOINTS, and CATEGORYRADIUS: the one-sided maximum has a closure algorithm. At every fixed gate $s_0 \geq 2$, Theorem 8.1 makes that maximum hard as well, restoring symmetry between the two extrema. Nonnegative nondecreasing weights are a sufficient tractability condition, not a necessary condition for polynomial MAX. Removing nonnegativity while keeping increasing weights includes the hard negative wedge.

For supplied CATEGORYRADIUS, provide a full-bag decomposition of each critical-cell summary; the polynomial number and size of these records preserve the reduction. The hard-wedge proofs establish the baseline category, crossing direction, and boundary attainment directly.

11. INTERPRETATION AND FURTHER QUESTIONS

The results are exact guarantees for the specified deterministic model. For nonnegative nondecreasing weights, the independent-box short outputs reduce to uniform offsets, or joint translation of all intensity thresholds. The independent box nevertheless has a richer full image, as Example 9.4 shows. General nonmonotone weighting introduces additional computational distinctions, including a polynomial maximum but hard minimum for positive decreasing pairs at gate one, and hardness of both extrema at every fixed gate at least two.

Threshold sensitivity and partial-volume effects in calcium scoring are established subjects [1, 13]. The present analysis supplies neither empirical calibration of a perturbation box nor a probability of coverage or clinical validity. Changes in graph adjacency, adaptive lesion definitions, and restrictions that exclude box corners require further analysis. These qualifications delimit application of the theorems; they do not identify common offsets with the entire independent perturbation family.

Several questions remain. The complexity of the parameter pair (width, g_0) with unbounded q is not settled here. The two-breakpoint positive-weight reduction uses input-dependent numerical weights; whether a fixed positive decreasing three-weight vector already yields hardness at gate one remains open. The explicit-image obstruction leaves output-sensitive enumeration and delay bounds open. More economical dynamic-programming states or additional graph restrictions may improve the positive parameter dependence. Finally, a complete formal treatment of the arbitrary-bitstring algorithms and their bit runtimes would extend the semantic formalization described in Appendix B.

Computational tools. OpenAI dots was used for mathematical development and computational checks with Astra; additional model-assisted reviews used GPT-6 Astra Pro and Claude Opus 5.5 max. Such reviews and finite computations support error detection but do not replace the proofs. The scope of the separate Lean formalization is specified in Appendix B.

APPENDIX A. CANONICAL ENCODINGS AND COMPLETE OUTPUT CONTRACTS

This appendix fixes the bit language used in Sections 6–8. An equivalent ordinary explicit binary representation would give the same complexity classifications, but the following contract makes validation and output claims unambiguous.

For $t \in \mathbb{N}$, let $N(t)$ be Elias gamma code of $t + 1$ and let

$$g(t) = 2\lfloor \log_2(t + 1) \rfloor + 1$$

be its exact length. For a reduced rational p/d , with $d \geq 1$, encode

$$R(p/d) = \text{sign}(p) \parallel N(|p|) \parallel N(d - 1), \quad R(p/d) = 1 + g(|p|) + g(d - 1).$$

The sign bit is one exactly for a negative numerator; zero is uniquely 0/1 with sign zero. Nonreduced representations and negative zero are invalid.

The numbered source body is the concatenation, in this order, of:

- (i) $N(n), N(|E|), N(q), N(s_0)$;
- (ii) $R(\tau)$ and the q breakpoint codes;
- (iii) the $q + 1$ weight codes;
- (iv) $(R(x_i), R(a_i))$ for $i = 0, \dots, n - 1$;
- (v) the edge pairs $(N(i), N(j))$, with $i < j$, in strictly increasing lexicographic order.

Every task begins with the literal header 10100001, the encoding version 00000011, a three-bit tag, and a one-bit mode. The tags 000, 001, 010, 011, 100, and 101 designate REACH, MAX, MIN, ENDPOINTS, CATEGORYRADIUS, and IMAGE, respectively. The tags 110, 111 are reserved and invalid. After the source body, REACH appends $R(e), R(K)$; each other fixed-radius task appends $R(e)$; and CATEGORYRADIUS appends $N(m)$ and the ordered category cutpoint codes. Mode zero has no further suffix. A selected task/mode algorithm treats a mismatched tag or mode as invalid.

A decomposition record contains $N(r)$, its $r - 1$ sorted tree-edge pairs, and then r bags, each a natural-coded length followed by its strictly increasing vertex list. It must satisfy all decomposition conditions of Section 7. A fixed-radius mode-one input appends one record for its constructed summary. A category mode-one input appends $N(2t)$ and one record for each canonical representative, in exactly their cell order. Empty summary graphs still require a nonempty decomposition tree. Trailing bits are invalid.

The source state code is

$$N(n) \parallel \big\|_{i=0}^{n-1} N(\text{code}(s_i)), \quad \text{code}(\perp) = 0, \quad \text{code}(a_j) = j + 1.$$

The relative parser checks the exact count, alphabet, and complete consumption. It tests syntax independently of box legality.

A function error is the single bit 0. Successful MAX and MIN outputs are $1\|\mathbf{R}(S_e^+)\|$ and $1\|\mathbf{R}(S_e^-)\|$; ENDPOINTS returns $1\|\mathbf{R}(S_e^-)\|\mathbf{R}(S_e^+)\|$. The category answer is 10 for infinity, and $11\|\mathbf{R}(d)\|v_d$ for a finite boundary. If the exact image is $r_1 < \dots < r_h$, its output is $1\|\mathbf{N}(h)\|\mathbf{R}(r_1)\|\dots\|\mathbf{R}(r_h)\|$. There is no padding or compressed substitute for this list.

A.1. Summary payload. Number uncertain nodes first in original vertex order, then stable components in order of their least source vertex. The payload concatenates q, s_0, u, k, h as natural codes; all weights; γ if and only if $u > 0$; each uncertain record (A_u, β_u) ; each component record $(t_C, A_C, \alpha_C, \beta_C)$; and sorted summary edges. Its exact bit length is

$$\begin{aligned}
B_\Sigma = & g(q) + g(s_0) + g(u) + g(k) + g(h) + \sum_{j=0}^q R(\lambda_j) + \mathbf{1}_{u>0}g(\gamma) \\
& + \sum_{u \in U} (R(A_u) + g(\beta_u)) \\
& + \sum_{C \in \mathcal{C}^+} (g(t_C) + R(A_C) + g(\alpha_C) + g(\beta_C)) \\
& + \sum_{(i,j) \in E(H_e), i < j} (g(i) + g(j)). \tag{A.1}
\end{aligned}$$

The conditional term is absent when $u = 0$; no value of γ is then used. If $a_v = p_v/d_v$, define

$$D_C = \prod_{v \in C} d_v, \quad P_C = \sum_{v \in C} p_v \prod_{w \in C \setminus \{v\}} d_w.$$

Since $A_C = P_C/D_C$ and reduction decreases numerator and denominator,

$$R(A_C) \leq 1 + g(P_C) + g(D_C - 1).$$

Formula (A.1) includes no hidden source supports, intensities, internal edges, or witness-lifting data.

APPENDIX B. SCOPE OF FORMAL VERIFICATION

The companion Lean project for version 5.5 formalizes the rational source model and the semantic results listed in Table 1. Its score is evaluated on the connected components of the actual active graph, using original vertex cardinalities, nonnegative rational areas, and maximum-bin weights. The general results allow arbitrary signed weights, zero areas, empty active sets, disconnected graphs, no density breakpoints, no category cutpoints, and gates larger than the graph. Nonnegative nondecreasing weights are additional hypotheses only for the monotonicity and corner results.

The summary is constructed from the radius-dependent stable and uncertain vertices. Its local domains, component correspondence, rational surjectivity, and exact score-image equality are proved for that construction. A separate summary payload contains only its graph, aggregate masses and areas, gate, weights, and finite local domains; its evaluator does not access source intensities or vertex supports. The radius is defined as the infimum of the actual bad rational radii in the extended nonnegative reals. Its critical-cell formula and independent boundary test are proved against that definition.

The table gives module names and representative local declaration names. Modules and declarations have the root namespace `ThresholdComponentScores`. Local declaration names below lie in its `Instance` namespace unless the row specifies another subnamespace. The release's `docs/formal-coverage.md` gives a fuller correspondence and the scope of the typed task specifications.

Table 1: Mathematical results and selected Lean declarations.

Result	Module and declarations
Lemma 3.2 and Theorem 3.3	Local (Thresholds): representative_spec; Score: liftState_spec, stateOf_surjOn, score_eq_stateScore, scoreImage_eq_stateImage
Corollary 3.4	Range: attained_extrema, scoreImage_has_endpoints, largest_absolute_error
Lemma 4.1	SummaryDomains (Thresholds): uncertain_allowed_iff, stable_max_bins_iff, stable_max_witness
Lemma 4.2	SummaryGraph (StableSummary): components_bijOn_expand, card_expand, sum_expand, sup_expand;
Theorem 4.3	SummaryPartition: stateScore_eq_contractedScore SummaryRealization: projectSummary_surjOn, projectSummary_stateOf_surjOn, scoreImage_eq_contractedImage, contractedScore_rational_lift, summary_attained_extrema; SummaryPayload: scoreImage_eq_payloadImage, payload_rational_lift
Lemmas 5.1 and 5.2	RadiusCells: scoreImage_cell_constant; Radius: categoryInvariant_iff_endpoints
Theorem 5.3	RadiusFormula: exists_radiusCell, categoryRadius_eq_min_badCriticalEndpoints, categoryRadius_eq_top_iff_no_bad_cells, finite_radius_and_boundary; Radius: categoryInvariant_zero
Proposition 9.1, semantic assertions	Monotone: score_mono; MonotoneCorners: corner_extrema, computeCornerReach_correct, computeCornerEndpoints_correct; MonotoneRadius: computeCornerRadius_correct, computeBoundaryInvariant_correct, computeCornerRadius_boundary_correct

The typed corner procedures are noncomputable semantic specifications on well-formed mathematical inputs. Their correctness establishes the meanings of threshold reachability, extrema, radius, and the separate boundary Boolean; it does not establish executable extraction, validation of arbitrary bitstrings, or polynomial running time. In particular, REACH means a score at least the input threshold.

The encoding and bit-length bounds, summary-size accounting, explicit-image output obstruction, complete validation and evaluation algorithms, costed dynamic program, and all classical and parameterized complexity classifications are paper proofs outside this formalization. The fixed-pair classification, closure construction, common-offset corollary, and summary-width filtration are likewise not claimed as Lean-verified results. Thus the companion project verifies the specified semantic portion of the article, not the entire manuscript.

Dependency pins are Lean 4.32.0 and Mathlib 81a5d257c8e410db227a6665ed08f64fea08e997. The associated release supplies the Lean sources, dependency manifest, and build instructions. Compiling this article is a separate operation from checking those proofs.

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